

Perception

Sensors

Objective

- Short introduction to perception
- Overview of the most commonly used sensors in Robotics.
- Understand their use and their basic operation.
- Understand how sensors are characterized, perform
- Understand that sensors are not perfect and propagates uncertainties

Autonomous mobile robot: the see-think-act cycle

Knowledge, database

Localization
Map Building

environment model
local map

Information
Extraction

Raw data

Sensing

Perception

- Takes measurements using various sensors(sensing)
- Then extract meaningful information from those measurement (information extraction)
- Extracted data can then be used for localization and human-robot interaction tasks

- Challenging: the nature of the extracted data rely on the environment and changes depending of it.
- Environment is very stochastic and uncertain
→ difficult to build a reliable model of it

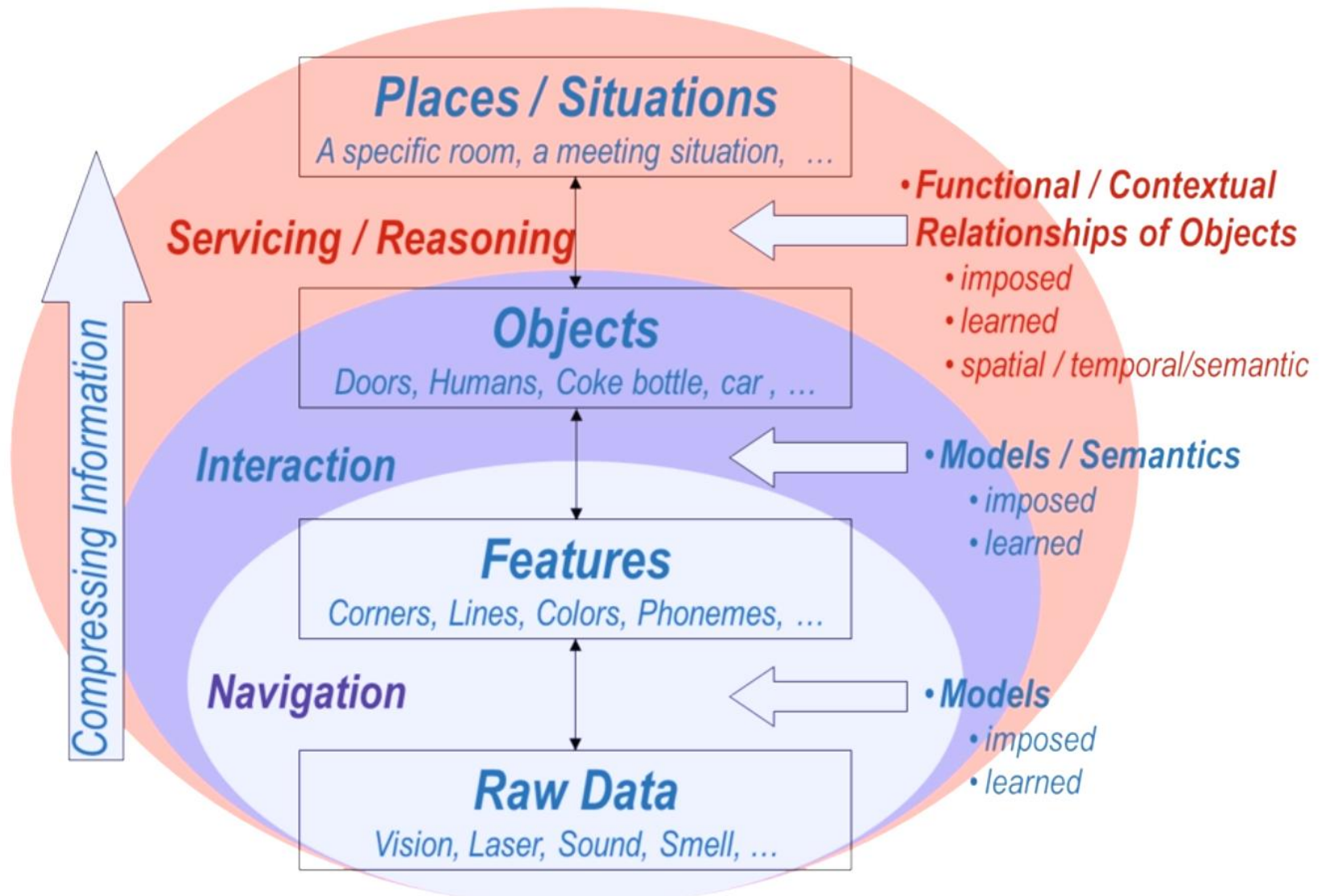
- Main task: building environment models so that the robot can use to localize and then navigate into.

Enviroment



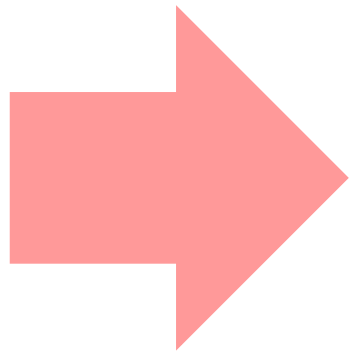
robotic application

Perception: definition



Perception: challenges

- Dealing with real-world situations
- Reasoning about a situation
- Cognitive systems have to interpret situations based on uncertain and only partially available information
- They need ways to learn functional and contextual information (semantics / understanding)

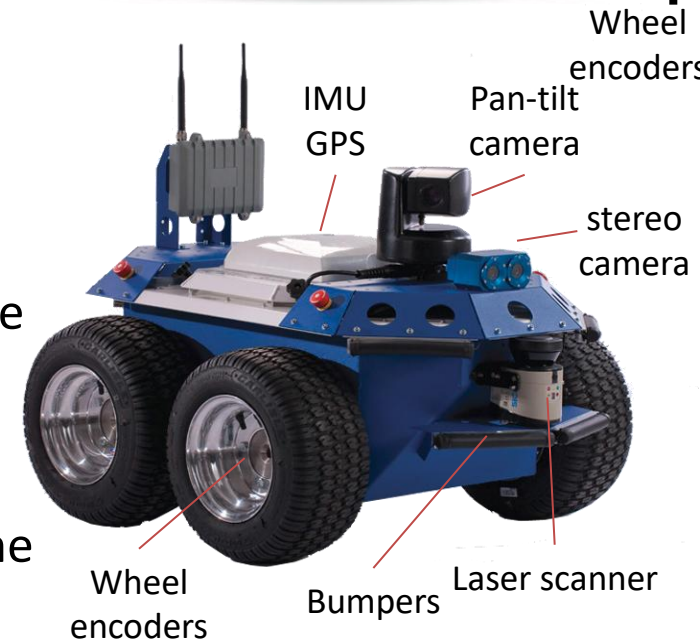
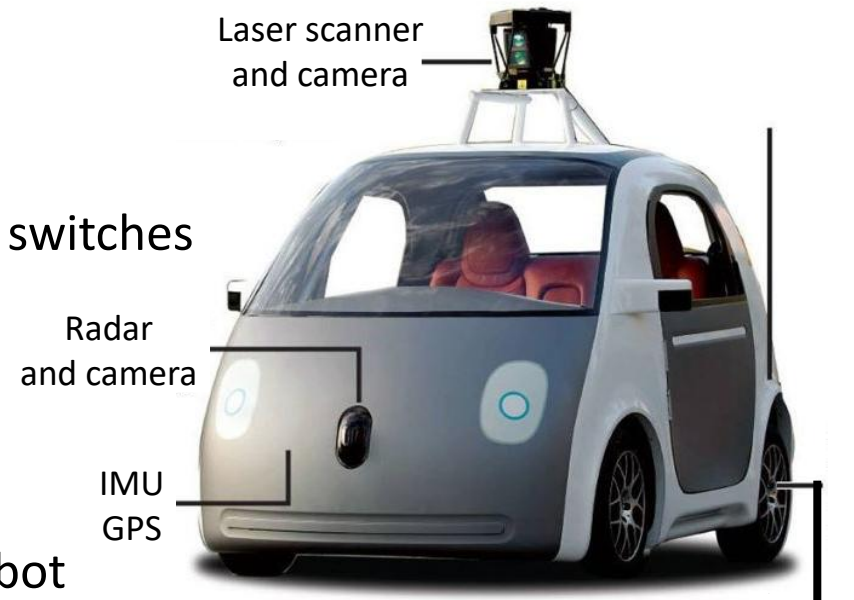


Probabilistic
Reasoning



Sensors: common sensors and their use in mobile robotics

- Tactile sensors or bumpers
 - Detection of physical contact, security switches
- GPS
 - Global localization and navigation
- Inertial Measurement Unit (IMU)
 - Orientation and acceleration of the robot
- Wheel encoders
 - Local motion estimation (odometry)
- Laser scanners
 - Obstacle avoidance, motion estimation, scene interpretation (road detection, pedestrians)
- Cameras
 - Texture information, motion estimation, scene interpretation



Classification of sensors

■ What:

■ Proprioceptive sensors

- Measure values internally to the system (robot)
(e.g. motor speed, wheel load, heading of the robot, battery status)

■ Exteroceptive sensors

- Information from the robots environment
- Distances to objects, ambient light intensity, env. features extraction

■ How:

■ Passive sensors

- Measuring energy coming from the env.; influenced by the environment

■ Active sensors

- Emit their proper energy and measure the reaction
- Better performance, but some influence on the environment

Classification of sensors

Sensor type	Sensor System	Proprioceptive (PC) or Exteroceptive (EC)	Active or Passive
Tactile sensors	Bumpers	EC	P
Wheel/motor sensors	Brush encoders	PC	P
	Optical encoders	PC	A
Heading sensors	Compass	EC	P
	Gyroscope	PC	P
	Inclinometer	EC	A/P
Acceleration sensors	Accelerometer	PC	P
Beacons	GPS	EC	P
	Radio, ultrasonic, reflective beacons	EC	A
Motion/speed sensors	Doppler: radar or sound	EC	A
Range sensors	Ultrasonic, laser rangefinder, structured light, time of flight	EC	A
Vision sensors	CCD/CMOS cameras	EC	P

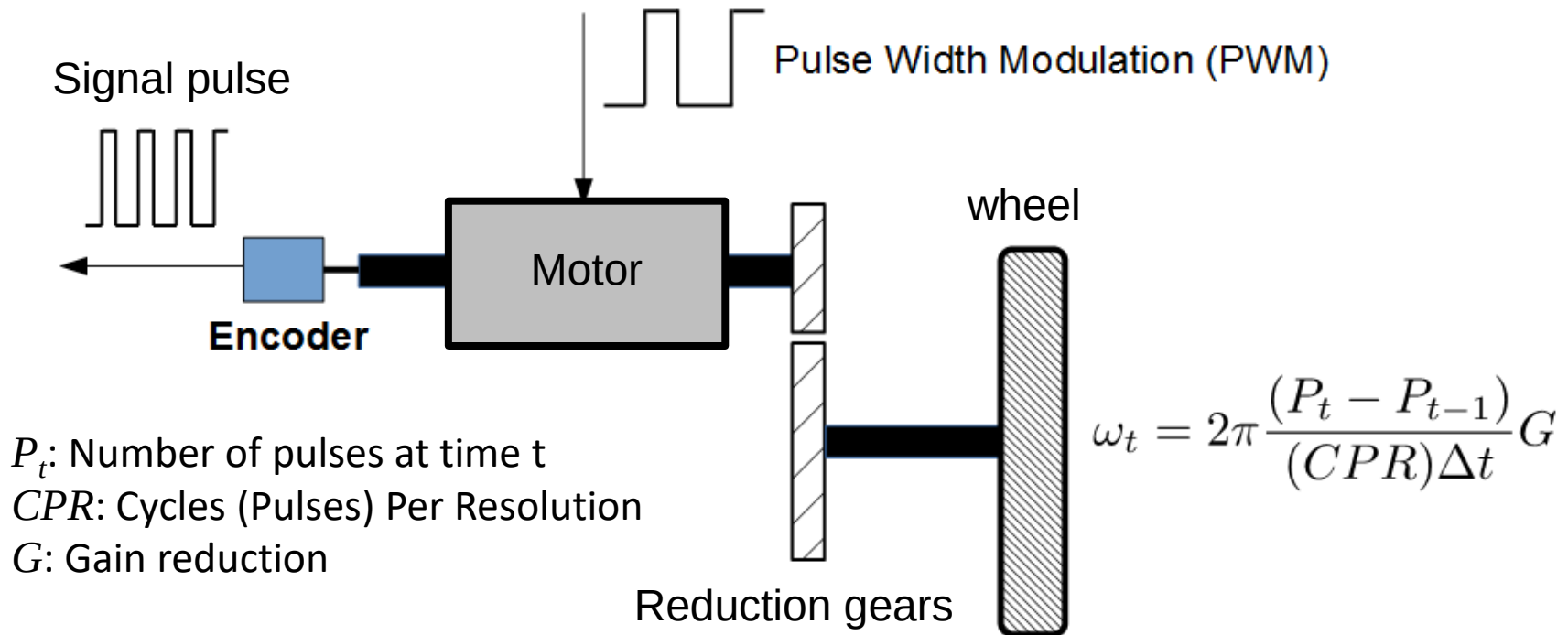
Encoders

- **Electro-mechanical devices** that convert linear or angular position of a shaft to an analog or digital signal, making it a linear/angular transducer



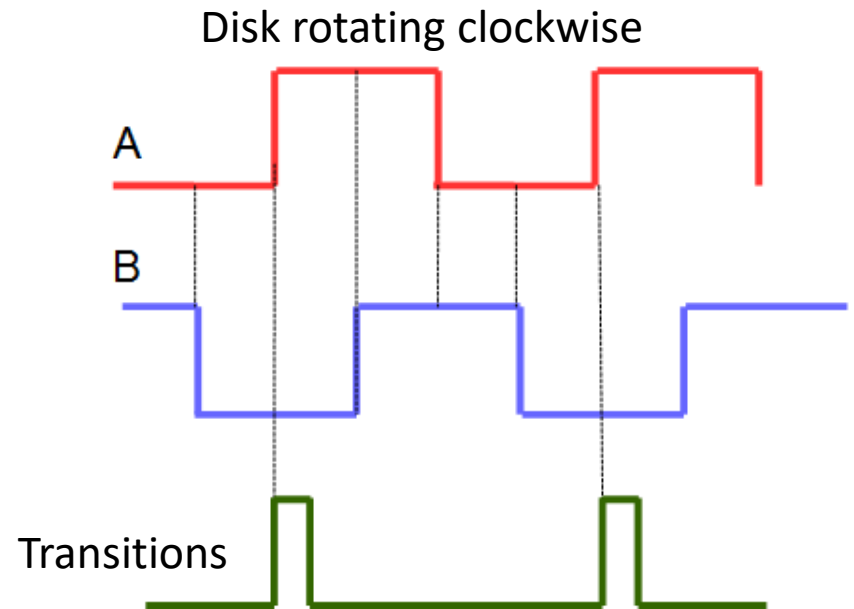
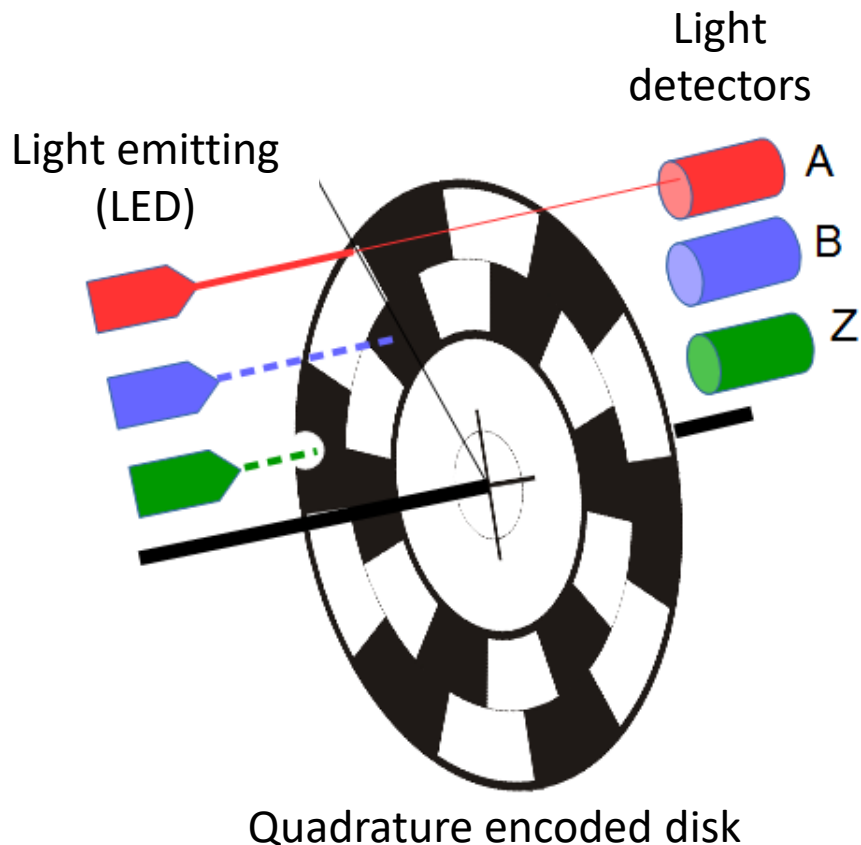
Encoders

- Encoders are angular position sensors (speed sensors are called tachometers). Encoders allow to measure rotation more accurately, for example, tenths of degrees / s.



Quadrature Encoders

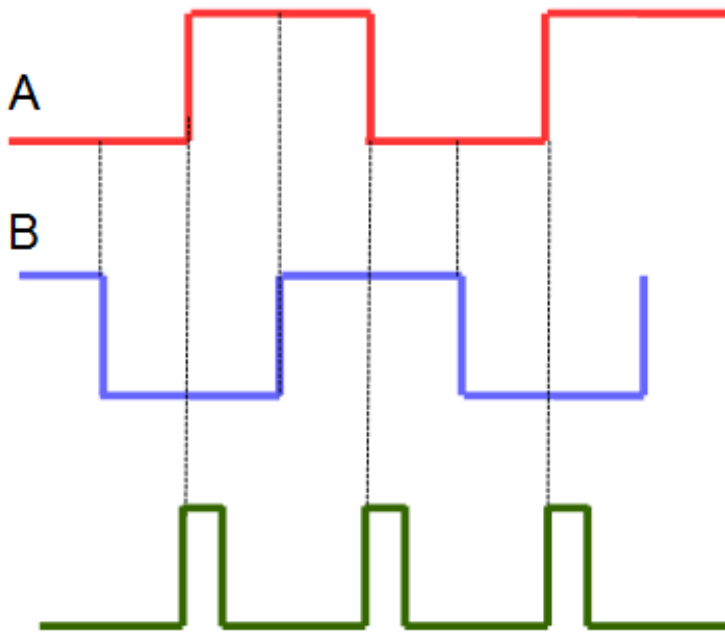
- A **quadrature encoder**, also known as an incremental rotary *encoder* measures the speed and direction of a rotating shaft. *Quadrature encoders* can use different types of sensors, optical and hall effect are both commonly used



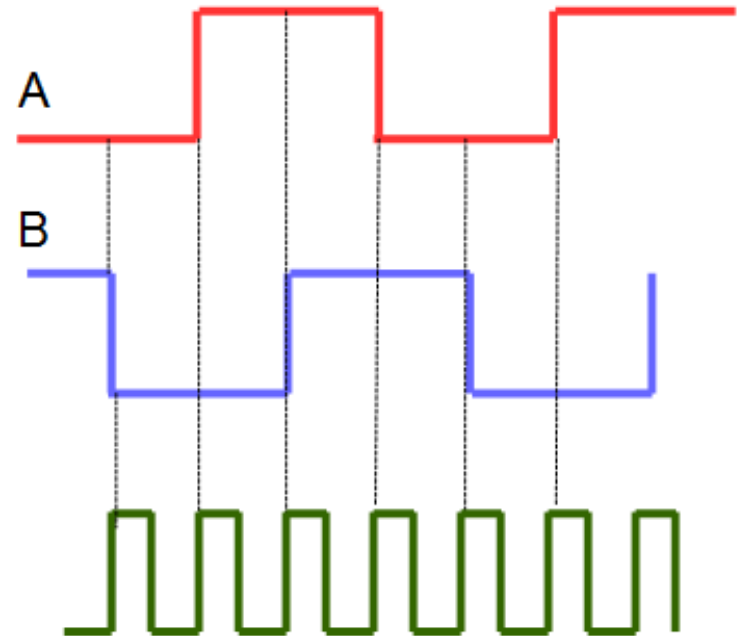
The transitions count measures the angular position of the disc while the relative phase of the channel dictates the direction of rotation. The Z channel provides a pulse every rotation.

Quadrature Encoders

- It is possible to increase the resolution of a quadrature encoder with the 2x and 4x encoding scheme.



Codification x2



Codification x4

Quadrature Encoders

- **Typical specs:**

CPR: 200-10000.

Supply and output: 5-24 V.

Consumption: 100 mA.

Maximum pulse rate: 100 KHz.



Cytron Encoder B-106
(US\$ 135,00)

- With a microcontrollers, the pulse count is given by counting interruptions caused by the transitions combined with the readings of level in the other channel.

The slip of the robot wheels on the floor “tricks” encoders.

Motors used in mobile robots often have integrated encoders on their axis.

Odometry

- Odometry is a measure of the robot's movements along time.
- The sum of movements provides the pose of the robot and the total traveled distance .
- The odometry is commonly computed by reading the encoders and feeding the kinematic model of the robot:

$$\begin{bmatrix} x \\ y \\ \theta \end{bmatrix}_t = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}_0 + \sum_{k=1}^t \begin{bmatrix} \Delta_s \cos(\theta + \frac{\Delta\theta}{2}) \\ \Delta_s \sin(\theta + \frac{\Delta\theta}{2}) \\ \Delta\theta \end{bmatrix}_k$$

$$\Delta_s = \frac{R(\omega_r + \omega_l)}{2} \Delta_t \quad \omega = \frac{R(\omega_r - \omega_l)}{2L} \quad \Delta\theta = \omega \Delta_t$$

Where ω_r and ω_l are given by the encoders

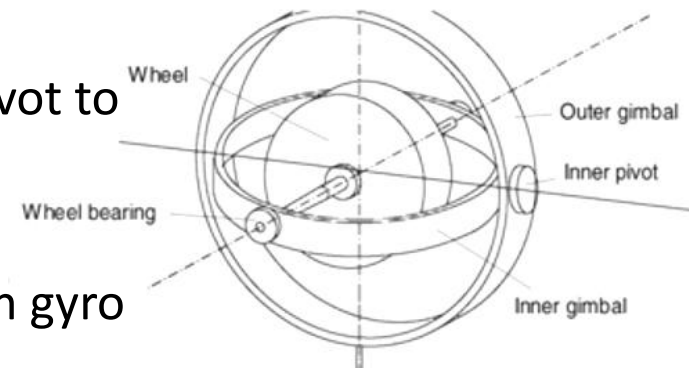
Heading sensors

- Definition:
 - Heading sensors are sensors that determine the robot's orientation and inclination with respect to a given reference
- Heading sensors can be proprioceptive (gyroscope, accelerometers) or exteroceptive (compass, inclinometer)
- Together with an appropriate velocity information, they allow integrating the movement to a position estimate.
 - This procedure is called **deduced reckoning** (ship navigation)
- Sensor types:
 - Compass: sense the absolute direction of Earth magnetic field
 - Gyroscope: senses the relative orientation of the robot with respect to a given reference.

Heading sensors: Gyroscopes

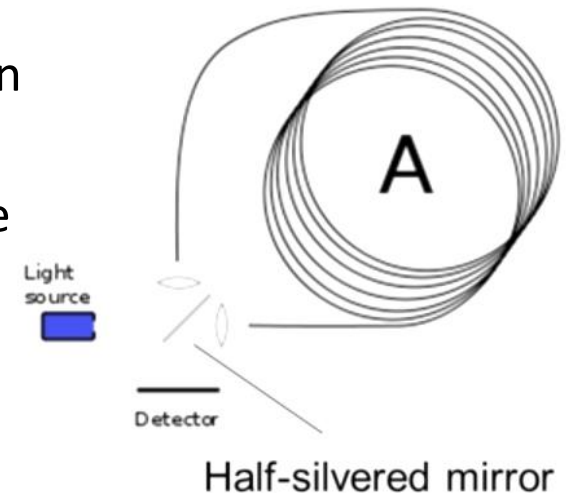
■ Mechanical gyroscopes:

- Angular momentum associated with a spinning wheel keeps the axis of the gyro inertially stable.
- No torque can be transmitted from the outer pivot to the wheel axis, however, friction in the axes generates drift
- Quality: 0.1° drift in 6 hours (a high quality mech gyro costs up to 100,000 \$)



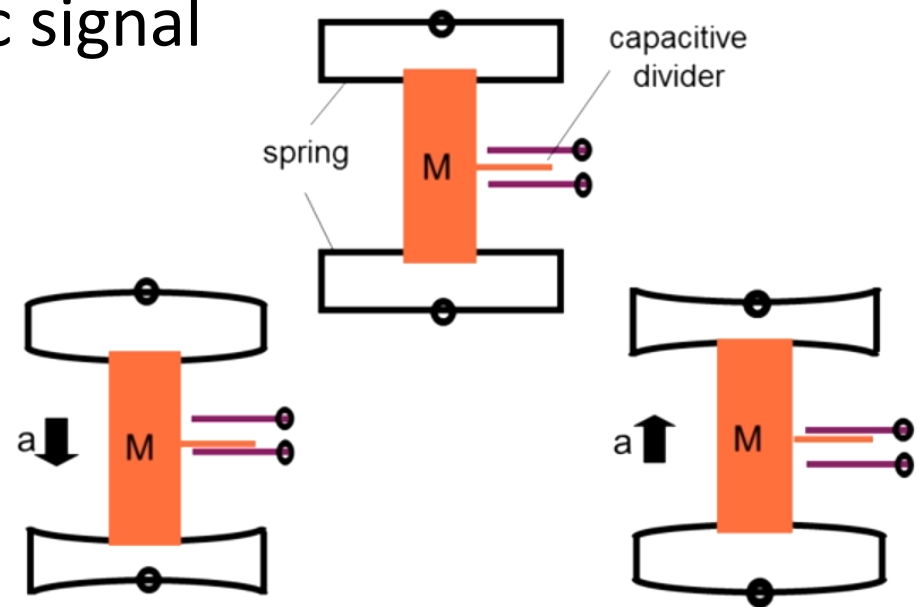
■ Optical gyroscopes

- Uses two monochromatic laser beams travelling in an optical fiber in two opposite directions
- The laser beam travelling in direction opposite to the rotation experiences a slightly shorter path (Sagnac effect)
- The phase shift of the two beams is proportional to the angular velocity.



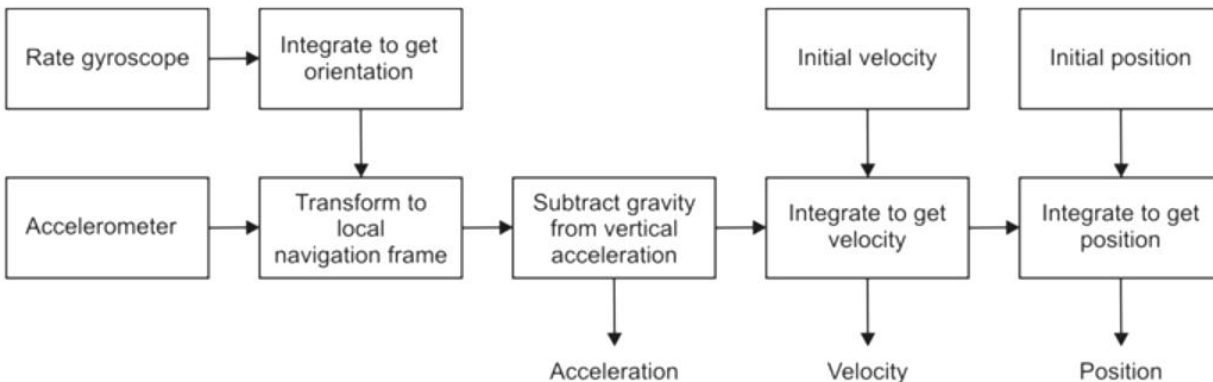
Heading sensors: MEMS accelerometer

- A spring like structure connects the device to a seismic mass vibrating in a capacity divider that converts the displacement into an electric signal
- Can be multidirectional
- Can sense up to 50 G
- Applications
 - Dynamic acceleration
 - Static acceleration (inclinometer)
 - Airbag sensors (+- 35 G)
 - Control of video games (e.g. Nintendo Wii) and HMI



Heading sensors: Inertial Measurement Unit (IMU)

- It uses gyroscope and accelerometer (6DoF IMU) to estimate the relative position (x,y,z), and orientation (roll, pitch yaw), velocity, and acceleration of a moving vehicle with respect to an inertial frame.
- 9 DoF IMUs have an additional 3DoF Compass using Hall effect so that the orientation is not relative but absolute.
- IMUs often have a temperature sensor for calibration.

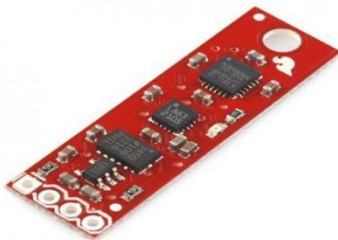


Demo IMU V-REP



Heading sensors: Inertial Measurement Unit (IMU)

- inertial measurement units are easily integrated with a microcontroller via I2C or SPI interfaces.
- The data provided must undergo a pre-calibration and a filtering:
 - Accelerometers are inherently imprecise
 - Gyroscopes are more accurate but suffer from drift with time that need external reference to be canceled.
 - magnetometers suffer the influence of magnetic fields generated by the robot itself or other equipment around. (e.g. our robotic wheelchair getting close to an extinguisher!!)
 - all sensors have your measurements influenced by internal temperature sensor, requiring their compensation.



Sparkfun Razor 9DoF IMU (US\$ 75.00)

Demo IMU V-REP



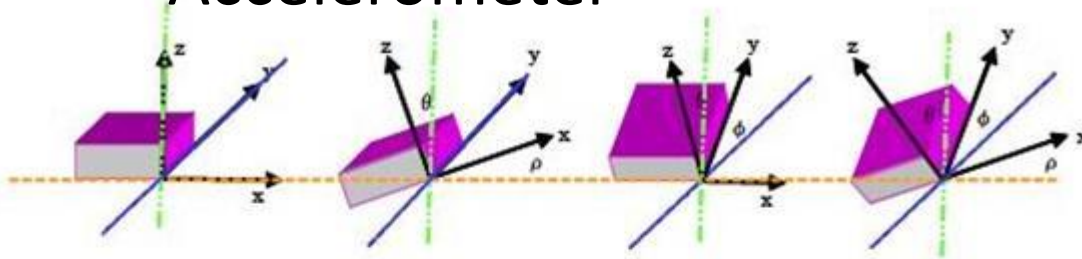
Analog Devices ADIS 16488
9DoF IMU (US\$ 1200.00)

Heading sensors: Inertial Measurement Unit (IMU)

- In mobile robotics gyroscopes are used to correct the orientation of the robot, which is usually computed by encoders that provide the rotation of the wheels (i.e. sensor fusion).
- The accelerometer is not often used to compute the robot's motion given its imprecision at low accelerations and errors due to double integration. It is however useful to measuring the robot's inclination (to prevent tipping over), using the gravity acceleration as reference.
- Aerial robots can combine the measurements of the gyroscope and accelerometer increasing the accuracy of their pose and attitude estimate as well as their control.
- The use of magnetometer is rarer given the influence of the magnetic fields produced by the robot or other electrical / metallic devices nearby. When used it must be positioned far from the base of the robot but still can be perturbed in an indoor environment.

Acceleration sensors

- Accelerometer

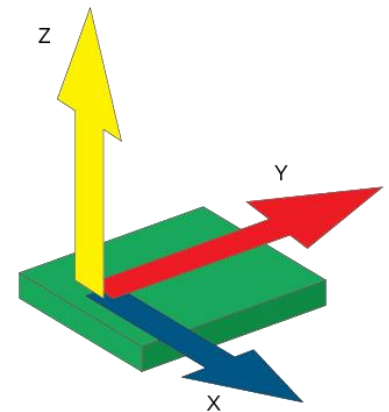
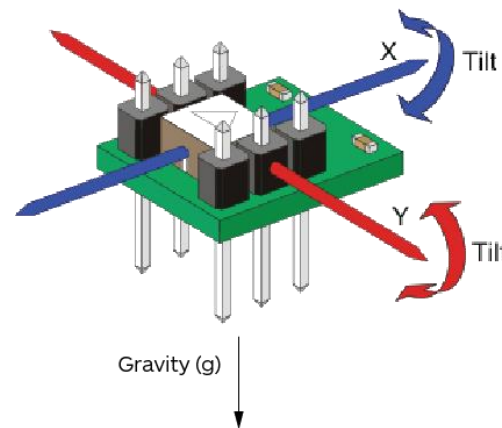


Three Axis for Measuring Tilt

$$\rho = \arctan\left(\frac{A_X}{\sqrt{A_Y^2 + A_Z^2}}\right)$$

$$\phi = \arctan\left(\frac{A_Y}{\sqrt{A_X^2 + A_Z^2}}\right)$$

$$\theta = \arctan\left(\frac{\sqrt{A_X^2 + A_Y^2}}{A_Z}\right)$$

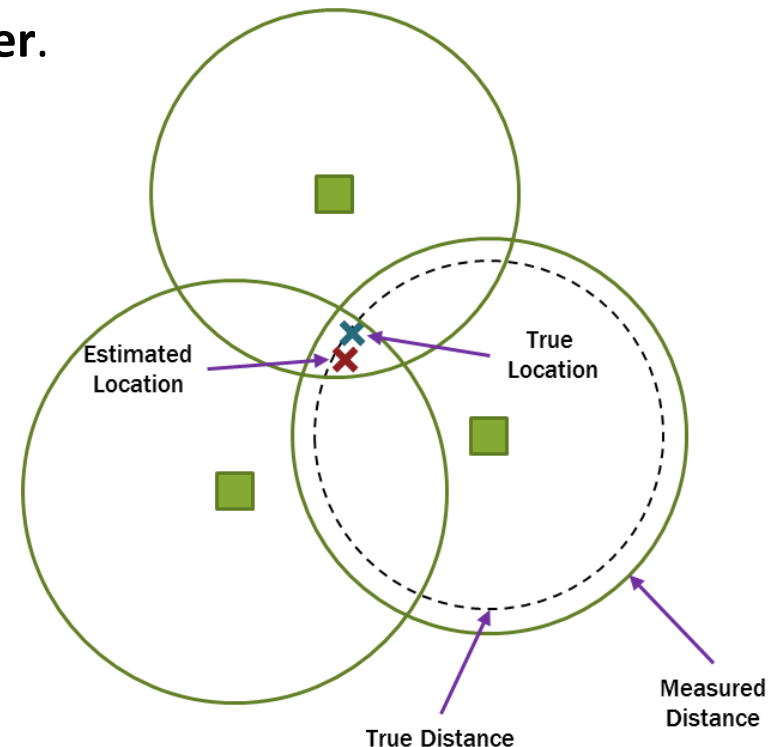


Demo esplora

Diagrams illustrating the axes of 2-axis (left) and 3-axis accelerometers. This particular 2-axis sensor is also capable of tilt measurement.
Image credit: Parallax | Kerry Wong

Beacons: GPS

- The **Global Positioning System** (GPS, aka Navstar) is a *global navigation satellite system* (GNSS) that provides location and time information in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites.
- Operates independently of any telephonic or internet reception (which can enhance the GPS positioning information).
- Freely accessible to anyone with a **GPS receiver**.
- Equivalent:
 - Russian: GLONASS
 - Indian: IRNSS
 - Chinese: BeiDou
 - European GALILEO



Beacons: GPS

- The United States government currently claims 4 meter RMS (7.8 meter 95% Confidence Interval) horizontal accuracy for civilian (SPS) GPS.
- Vertical accuracy is worse.
- Some devices/locations reliably (95% of the time or better) can get 3 or 2 meter accuracy.

* horizontal only



Autonomous < 15 m

Real-time < 1-5 m

Post-processed code phase differential 0.5 – 5 m

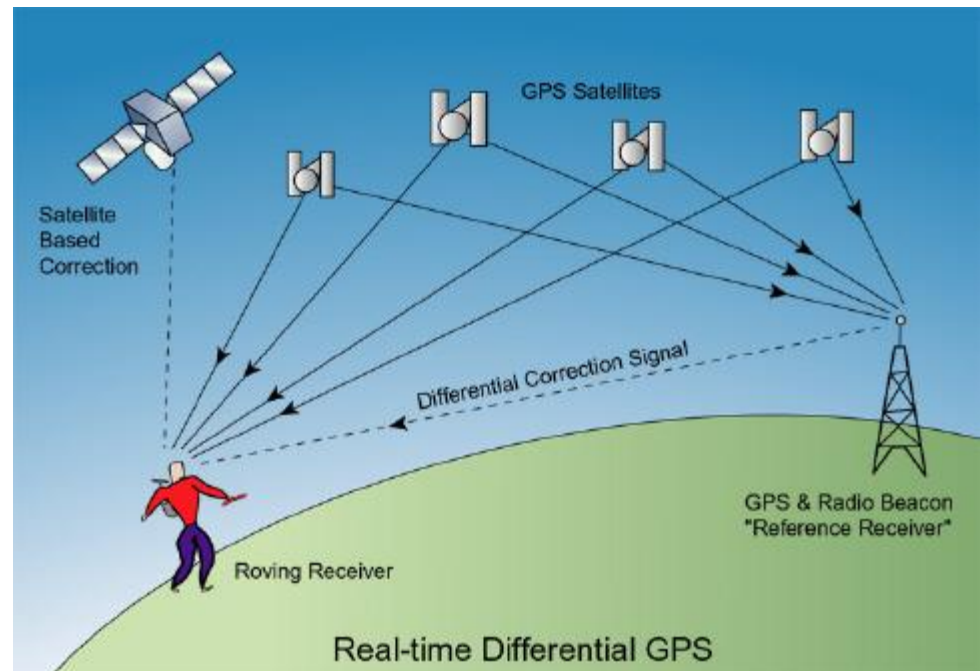
H-Star Processing < 0.2 – 0.3 m

Post-processed carrier phase differential 0.01 – 0.3 m



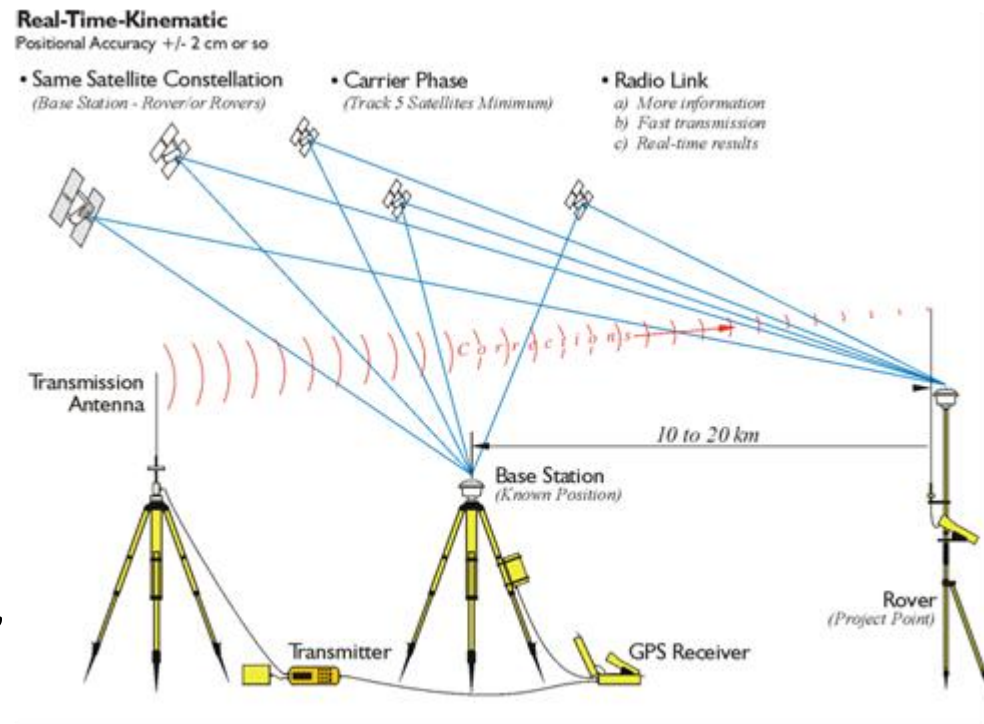
Beacons: DGPS

- Differential **Global Positioning System (DGPS)** is an enhancement to **Global Positioning System** that provides improved location accuracy, from the 15-meter nominal **GPS** accuracy to about 10 cm in case of the best implementations



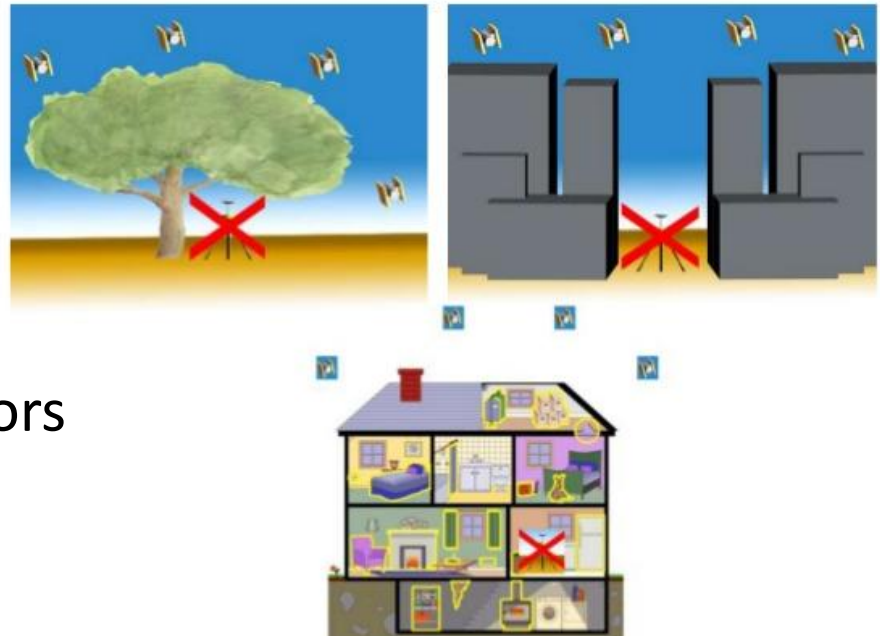
Beacons: RTK GPS

- **Real Time Kinematic (RTK)** satellite navigation is a technique used to enhance the precision of position data derived from satellite-based positioning systems GPS, GLONASS, Galileo, and BeiDou.
- Uses measurements of the phase of the signal's carrier wave, rather than the information content of the signal, and relies on a single reference station or interpolated virtual station to provide real-time corrections, providing up to centimeter-level accuracy.



Beacons: GPS

- **GPS denied environments:** places where the GPS does not work
- Indoor environments
- Covered environments
- Need to rely on other sensors or Odometry

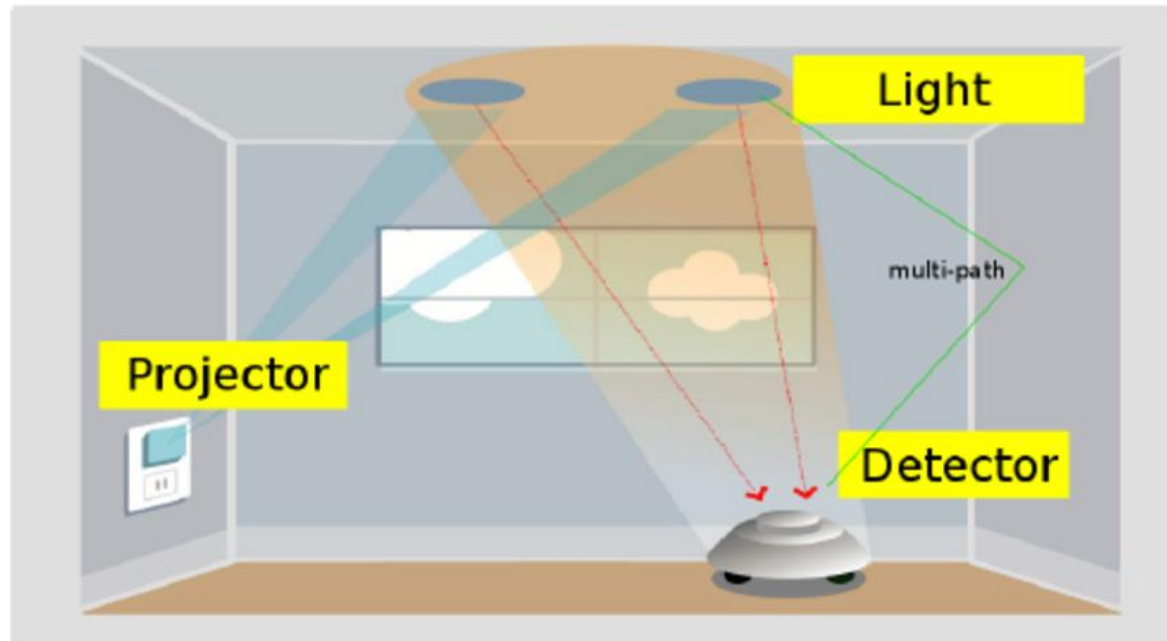


Beacons: Indoor Positioning System (IPS)

- System to locate objects or people inside a building using radio waves, magnetic fields, acoustic signals, or other sensory information collected by mobile devices.
- use technologies like distance measurement to nearby anchor nodes (nodes with known positions, e.g., Wifi access points), magnetic positioning, dead reckoning.
- They either actively locate mobile devices and tags or provide ambient location or environmental context for devices to get sensed.

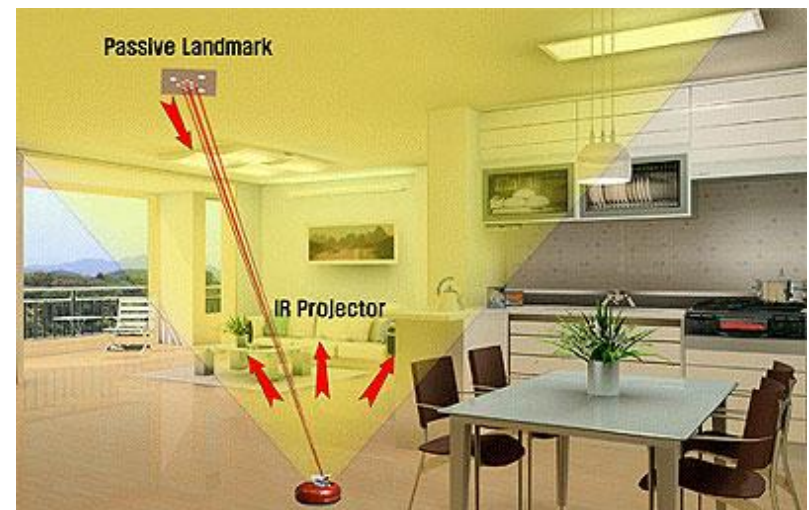
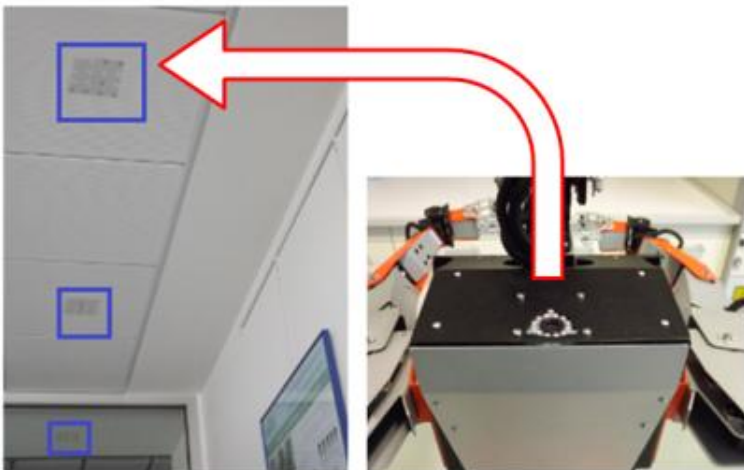
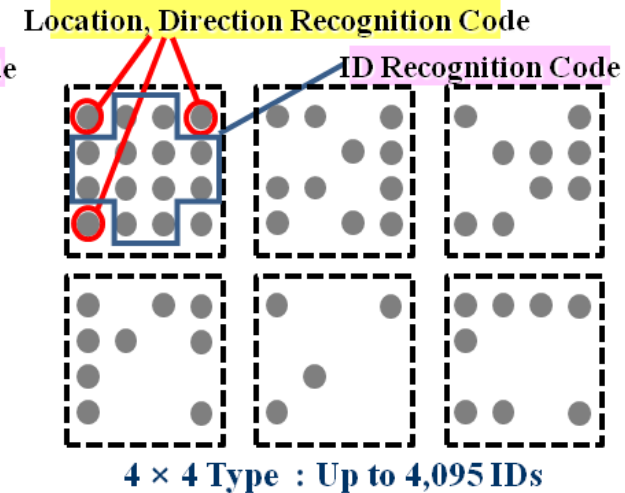
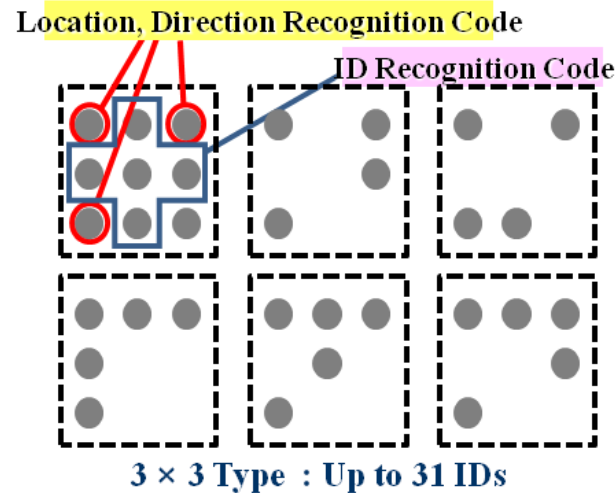
[video](#)

- **Northstar, Truetrack:**
 - _ Provides x, y, and θ with an acceptable accuracy
 - _ Requires plenty of beacons to cover the ground.



Beacons: Indoor Positioning System (IPS)

- **StarGazer**
- More accurate than northStar
- Requires plenty of tags
- Only one active source (embedded to the robot)
- Use the robot's battery

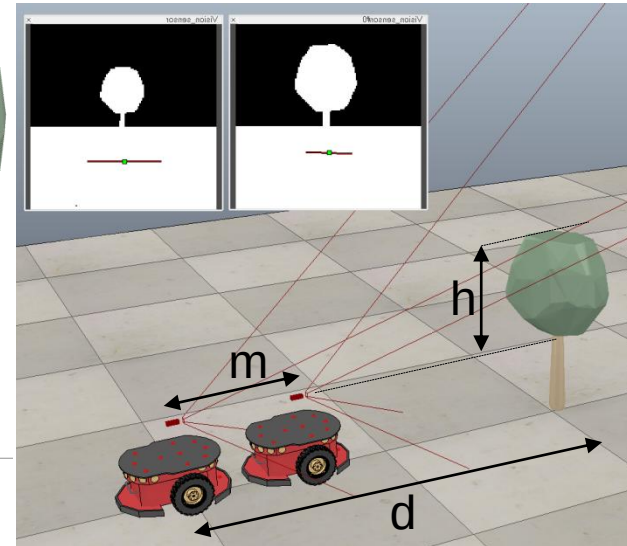
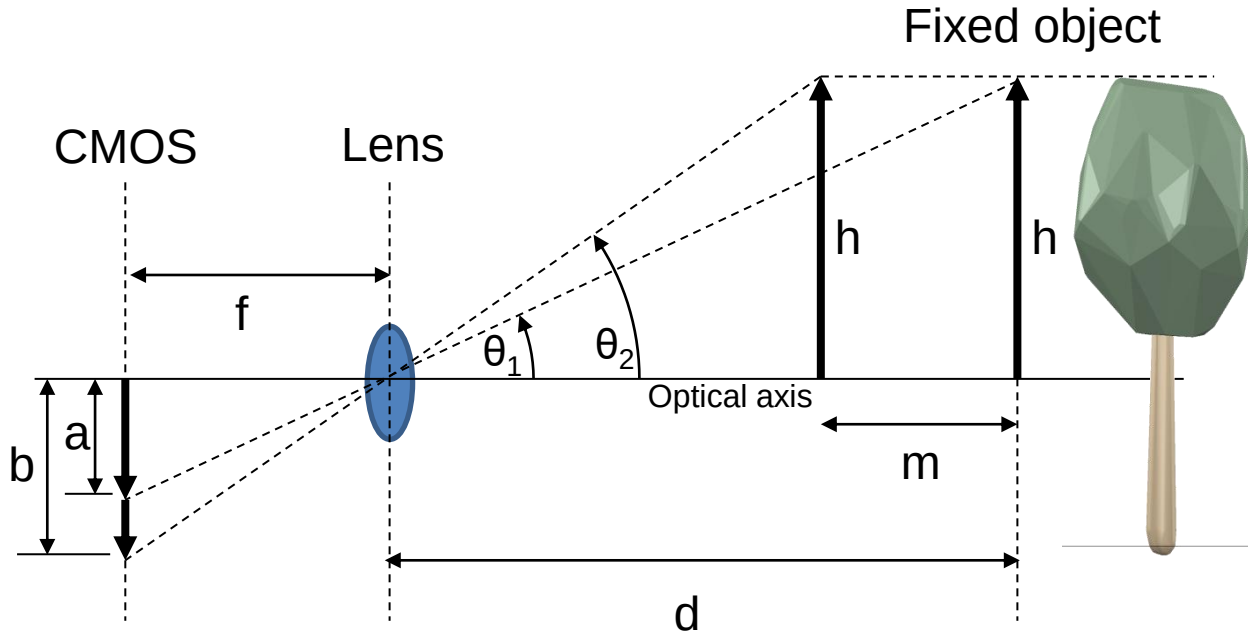


Cameras

- Cameras are passive sensors (i.e. not emit energy in the environment) and therefore consume little power.
- The interpretation of the captured image requires high processing ability using complex vision algorithms (can be done in dedicated hardware, such as FPGA, DSP and GPU).
- The image processing and thus the sensor reading is influenced by ambient light.
- Measurement of distance can be done using stereo vision (arrangement with 2 cameras), monocular camera with motion estimator, monocular camera with marker, or monocular camera with a known height and angle assuming flat ground (i.e. [point to go](#), [distance estimator with lane detection](#))
- There are many software packages for stereo vision, for example:
OpenCV: <http://opencv.org>.
Matlab Computer Vision Toolbox: <http://www.mathworks.com>.
LabVIEW 3D Imaging: <http://www.ni.com>.
SDKs, APIs and device drivers manufacturers (Point Gray, Focus Robotics, etc.).

Cameras

- Simple example of monocular camera distance estimation



$$\frac{a}{f} = \tan \theta_1 = \frac{h}{d} \quad \text{and} \quad \frac{b}{f} = \tan \theta_2 = \frac{h}{d-m}$$

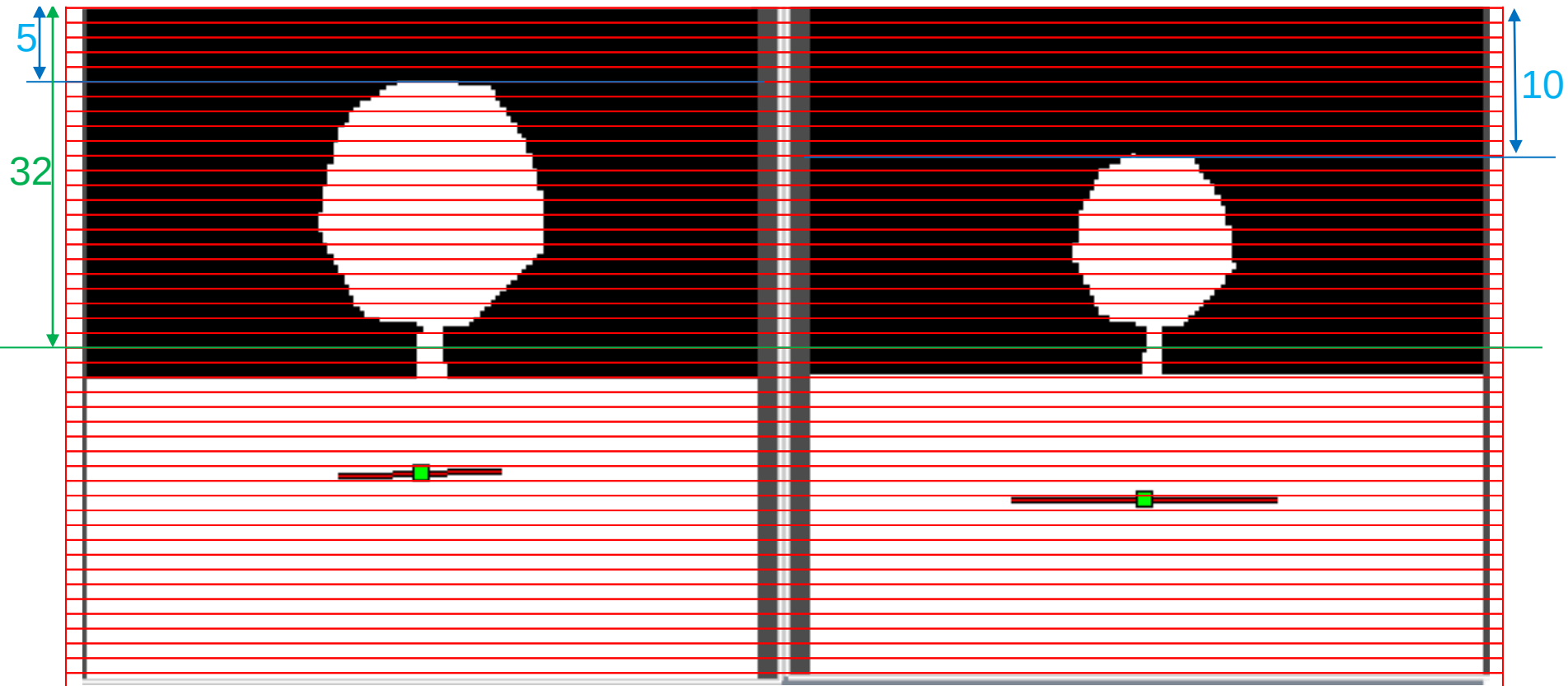
$$\frac{a}{b} = \frac{h}{d} \times \frac{d-m}{h} = \frac{d-m}{d} = 1 - \frac{m}{d}$$

$$\frac{m}{d} = 1 - \frac{a}{b} \quad \Leftrightarrow \quad d = \frac{m}{1 - \frac{a}{b}}$$

- h : height between optical axis and top of the object
- m : distance travelled by the robot between two frame acquisition
- d : distance to object at time $t-1$
- $d-m$ = distance at time t

Cameras

- Simple example of monocular camera distance estimation



Sensor resolution = 64x64

$a = 32 - 5 = 27$ pix

$b = 32 - 10 = 22$ pix

$m = 0.3$ m

$d_{\text{estimated}} = 1.08$ m

$d_{\text{measured}} = 0.96$ m

error $\approx 10\%$

Cameras: smart camera

- DVS128 Dynamic Vision Sensor

Dynamic vision sensors (DVS) are a completely new way of doing machine vision. Conventional video cameras use enormously redundant frames, wasting energy, computational power and time. The DVS uses patented technology that works like your own retina: only local pixel-level changes caused by moving in a scene are transmitted. The results are dramatic improvements in speed, power, dynamic range, data storage and computational requirements.



Demo V-REP and [Video](#)

Cameras: smart camera

- Pixy is a fast vision sensor you can quickly “teach” to find objects, and it connects directly to Arduino and other controllers.
- Robot vision made easy - press the button to teach Pixy an object.
- Simplify your programming - receive just the objects you're interested in.
- Use whatever controller you want - includes software libraries for Arduino, Raspberry Pi, and BeagleBone Black.
- Communicates via one of several interfaces: SPI, I2C, UART, USB or analog/digital output
- US\$ 70



[Video](#)

Cameras: stereo cameras

- **Typical specifications:**

Range and accuracy: depending on the image size and arrangement of the cameras.

Image Size: 640x480 (VGA), 752x480 (WVGA), 1024x768 (XVGA).

Image Type: Monochrome

Sensor: CMOS

Capture rate: 10-60 fps.

Consumption: <5 W

Connectivity: USB, FireWire, Ethernet, NTSC.



Focus Robotics
nDepth with FPGA
(US\$ 4.000,00)



e-com Systems Capella
(US\$ 1.300,00)



Point Gray Bumblebee 2
(US\$ 1.900,00 VGA,
2.400,00 XVGA)



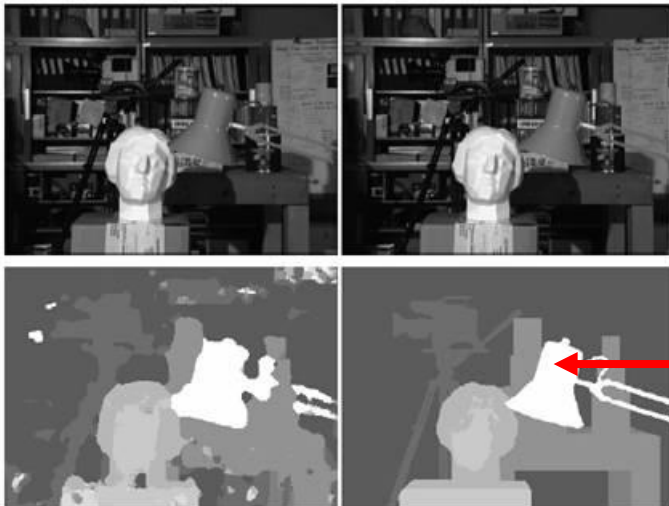
Point Gray Bumblebee XB3
(US\$ 3.500,00)

Cameras: stereo cameras

- **Fundamental principle:**

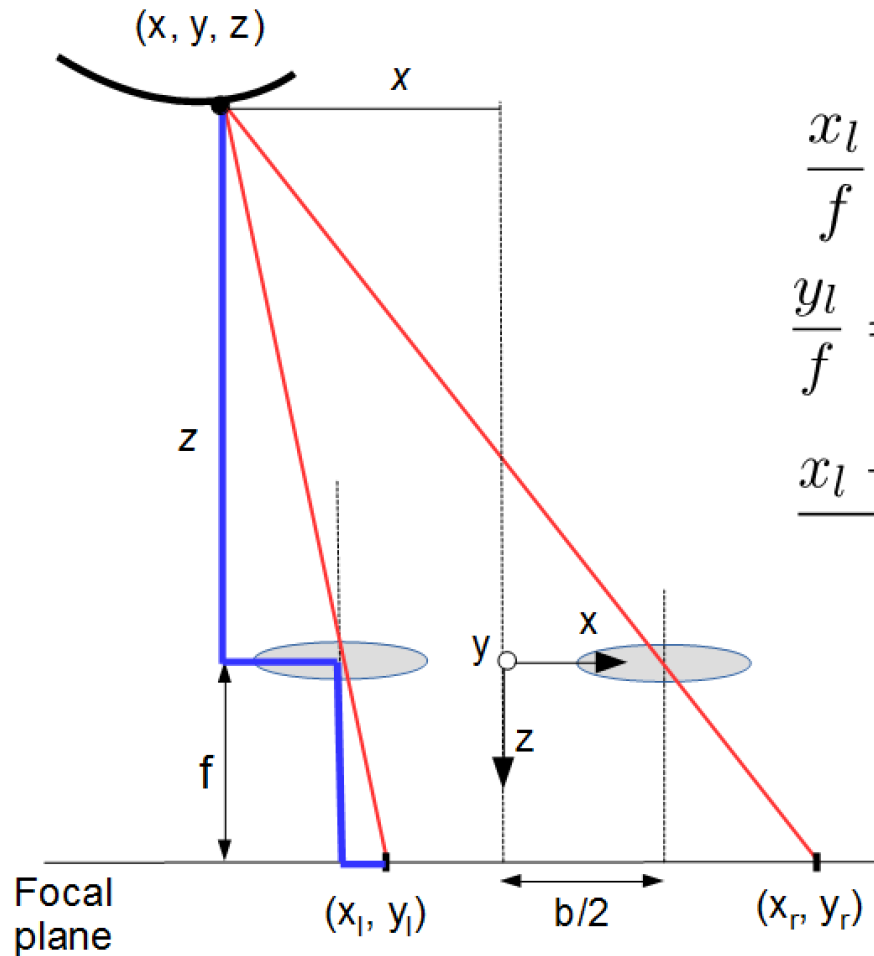
Given two images captured "simultaneously" by a stereo camera and a point in the scene identified in the two images, what is the distance from the camera to the point?

The processing of the two images produces images called disparity maps (disparity map) or depth (depth map). Each pixel in these images have their value proportional to the distance from the camera to the point in the scene given by the pixel.



Nearby points have greater pixel value (more white).

Cameras: stereo cameras



f and b are constants

$$\frac{x_l}{f} = \frac{x + \frac{b}{2}}{z}$$

$$\frac{y_l}{f} = \frac{y_r}{f} = \frac{y}{z}$$

$$\frac{x_l - x_r}{f} = \frac{b}{z}$$

$$\frac{x_r}{f} = \frac{x - \frac{b}{2}}{z}$$

b : baseline

$x_l - x_r$: disparity

$$x = b \frac{(x_l + x_r)/2}{x_l - x_r}$$

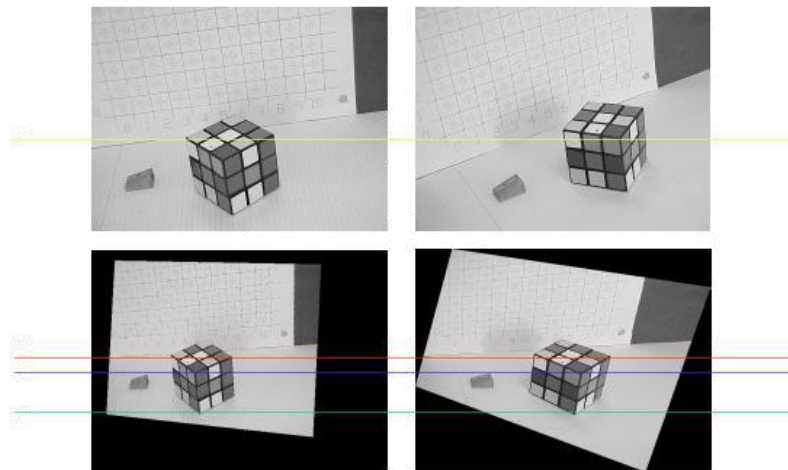
$$y = b \frac{(y_l + y_r)/2}{x_l - x_r}$$

$$z = b \frac{f}{x_l - x_r}$$

Cameras: stereo cameras

- Previous deductions assume:
 - without distortion lenses;
 - perfectly aligned identical cameras.
- As this does not happen in practice a calibration process of the cameras to compensate its misalignment and distortion of the lens is required.
- The result of this process is a set of parameters used to determinate the coordinates (x, y, z) .
- Stereo vision software packages provide calibration algorithms that use a chess frame as a reference image.

Image rectification (example)



Epipolar lines

Prof. Clésio Tozzi

Cameras: stereo cameras

- Finally, how the point (x, y, z) is identified in the two images?
It exists various algorithms such as SAD (Sum of Absolute Difference). These algorithms are also available in the stereo vision software packages.



- A matrix (usually 7×7 pixels) on the left image is defined, and moves a matrix of the same size, in the right image, along the corresponding epipolar line. The sums of absolute differences between pixels in the two matrices are compared. The point that produces the smallest difference is taken as corresponding point in two images.

Cameras: stereo cameras

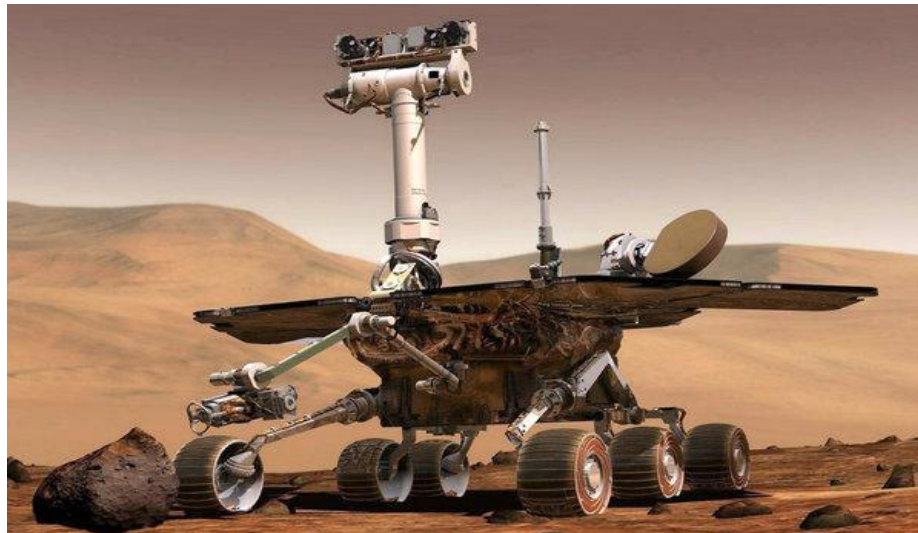
- SAD example (don't cry)
- This example uses the SAD differences to identify which part of a search image is most similar to a template image.
- The template image is 3 by 3 pixels in size, the search image is 3 by 5 pixels in size. Each pixel is represented by a single integer from 0 to 9.

template	Search Image	left	center	right																																																			
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3	1	1																																																					
3	3	1																																																					
0	2	0																																																					
1	3	4																																																					

- Three unique locations within the search image where the template may fit: the **left side**, the **center**, and the **right side** of the image.
- SAD values= absolute value of the difference between each corresponding pair of pixels
 $2 - 2 = 0$, $4 - 1 = 3$, $7 - 8 = 1$,....
- For each of these three image patches, the 9 absolute differences are added together, giving SAD values of 20, 25, and 17, respectively.
- From these SAD values: the right side of the search image is the most similar to the template image, because it has the lowest sum of absolute differences as compared to the other two locations.

Cameras: stereo cameras

- Stereo vision requires high processing capacity, possibly using dedicated hardware.
- Stereo vision requires images with great diversity of pixels. For example it is impossible to determine the distance to a wall with uniform color from stereo vision.
- However, in some scenarios, stereo vision is very adequate solution:



Range sensors

- Sonars.
- Laser Range Finder (LRF)
- Time of Flight (TOF) Cameras
- Structured Light



Sonar



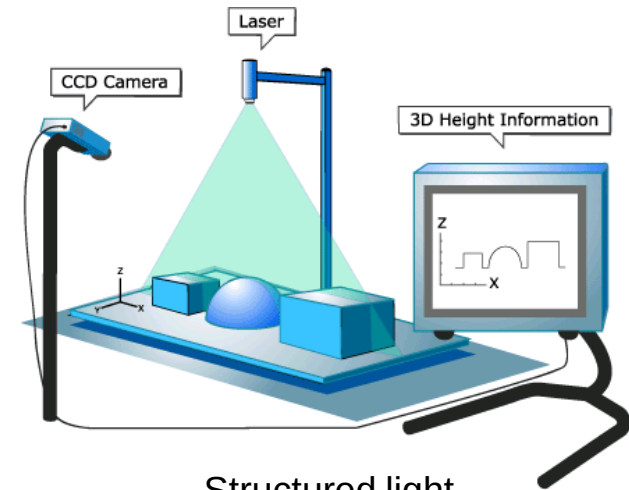
Laser Range Finder



SwissRanger (time of flight)



Kinect 2 (time of flight)



Structured light



[HDI 3D Structured-Light Scanners](#)



Kinect 1 (structured light)

Range sensors: Time of flight

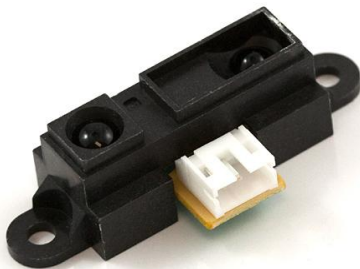
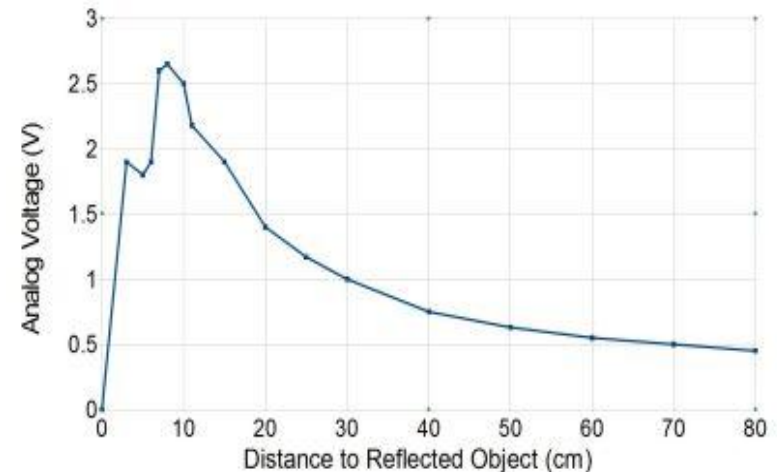
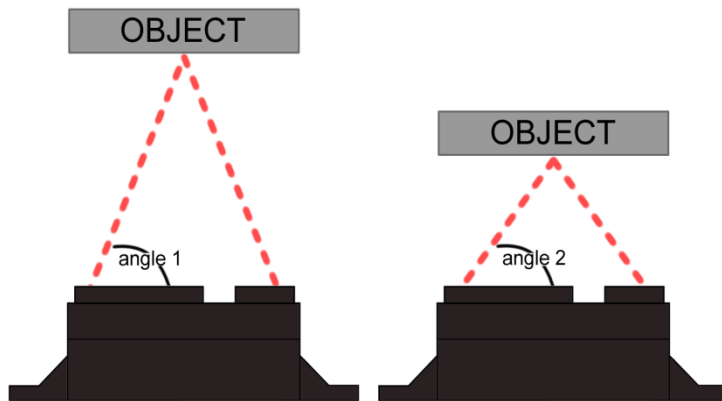
- Large range distance measurement -> thus called range sensors
- Range information:
 - Key element for localization and environment modeling
- Ultrasonic sensors as well as laser range sensors make use of propagation speed of sound or electromagnetic waves respectively
- The travel distance of a sound or electromagnetic wave is given by

$$d=c \cdot t$$

- d = distance travelled (usually round-trip)
- c =speed of wave propagation
- t = time of flight

Range sensors: Infrared Sensors

- Infrared light sensors measure distances, emitting light in the infrared range (~ 900 nm) and capturing the reflection. The intensity is captured by a sensor which produces an output dependent on the angle of incidence of the light.



Sharp PSD sensor (US\$ 25,00)

Arduino - V-REP demo

Range sensors: Infrared Sensors

- **Typical specs:**

Measurement range: 10 to 100 cm.

Inaccuracy: 5-10 mm.

wavelength: 900 nm.

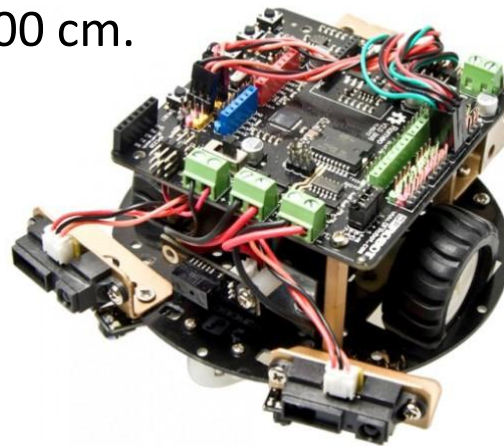
Opening angle: 20 degrees.

Scan rate: 40 readings / s.

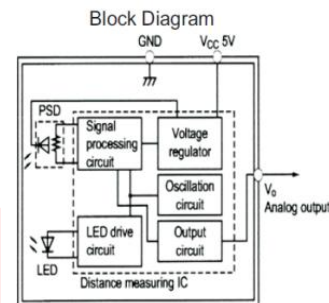
Power: +5 V

Consumption: 30 - 50 mA.

Connectivity: analog voltage.



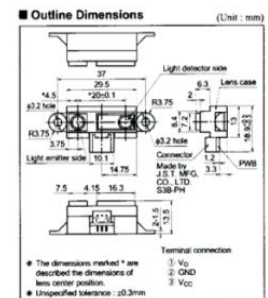
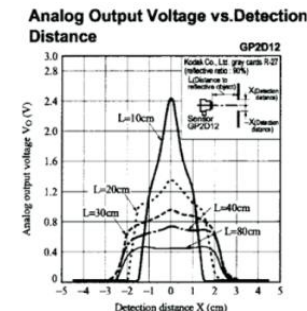
Operating Voltage	5 Vdc
Stand By Current	350μA
Output Signal	0-2.5 V DC
Dimension	19 x 35 x 14 mm
Detection Angle	10°
Detection Distance	10-80 cm



- Infrared light sensors rely on the ability of objects to reflect light at this wavelength. They have the same problems of sonar reflection.

- Their applicability are similar to sonar, but with less detection distance

V-REP modeling demo



Range sensors: Ultrasonic sensors

- Also known as sonars.
- Send sound pulses in the ultrasound range (above 20 KHz) and measure the return time of the echo (Δt).
- As the speed of sound in air is approximately constant ($v = 343 \text{ m / s}$), you can determine the distance of the object where the reflection is given:
$$d = v \cdot \Delta t / 2.$$



HC-SR04 (US\$ 4,00)



Parallax PING (US\$ 30,00)



Maxbotix HRXL (US\$ 100,00)

- The transmitter and receiver can be separate or integrated in the same device.
- The transmission and reception of sound is given by piezoelectric effect (mechanical vibration caused by electric field and vice versa).

Range sensors: Ultrasonic sensors

- **Typical specs:**

Measuring range: 5 cm to 5 m.

Inaccuracy: 5-20 mm.

Frequency: 40 - 80 kHz.

Pulse duration: 200 μ s.

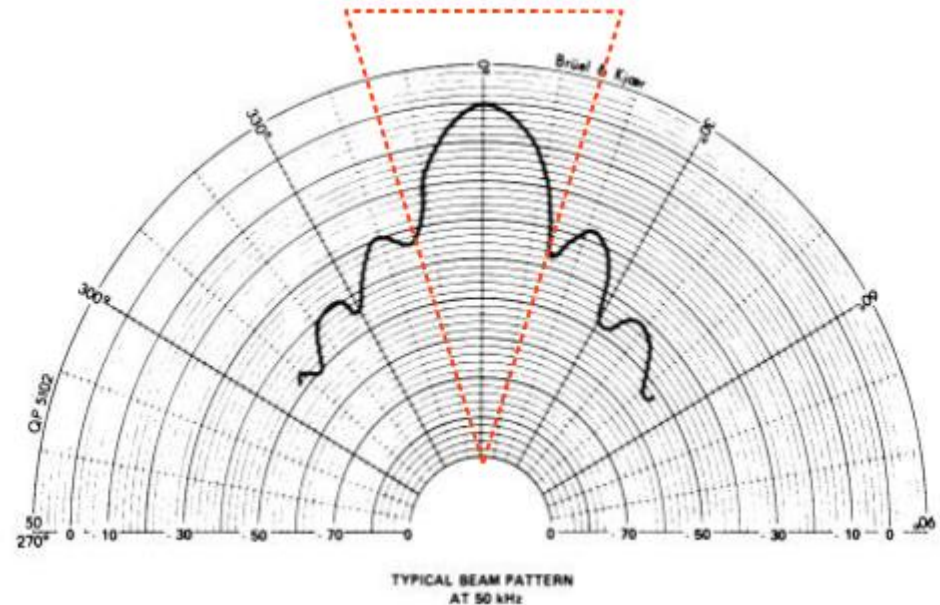
Opening angle 10 to 30 °

Scan rate: 50 readings / s.

Power: +5 V

Consumption: 20 mA.

Connectivity: TTL, I2C, RS-232.



Pros:

- Low cost.
- Size, reduced weight and consumption.
- Easy connection to microcontrollers.

Cons:

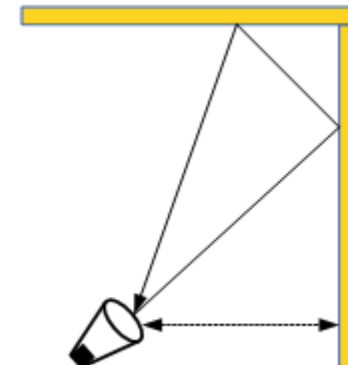
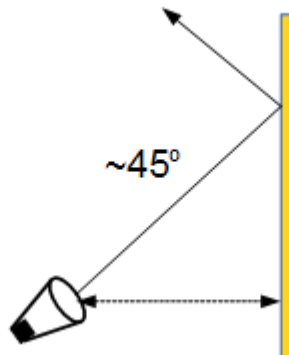
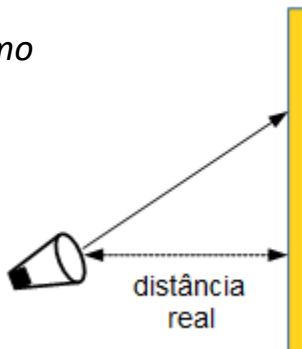
- Measured in a cone (not in a row).
- Interference between sensors.
- Unwanted reflections and absorbing materials interfere with the measurement.

Range sensors: Ultrasonic sensors

- As sonar detect objects within a cone with opening around 15-30 degrees, the precise position of the object is not known. Therefore it is common to use a "sonar ring".
- The problem is that the sonars can not operate simultaneously as their emission would interfere with the reading of the others (who's wave is who's?)
- Sequential operation is required and this reduces the read rate in a given direction (50 readings/s = i.e. every 20 ms: $20\text{ms} \times 343 \text{ m/s} = 6.86 \text{ m}$ max range)
- Another problem is the sound's reflection



V-REP demo



Range sensors: Ultrasonic sensors

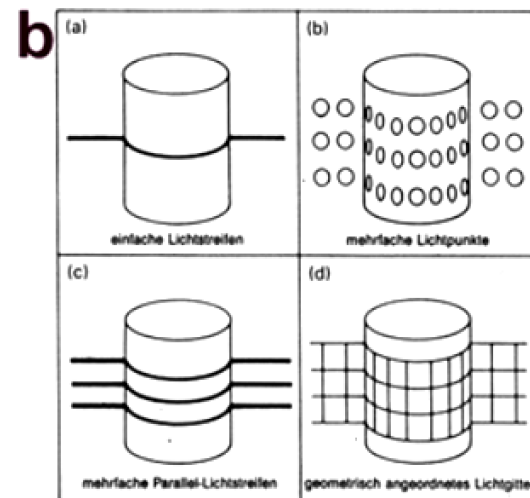
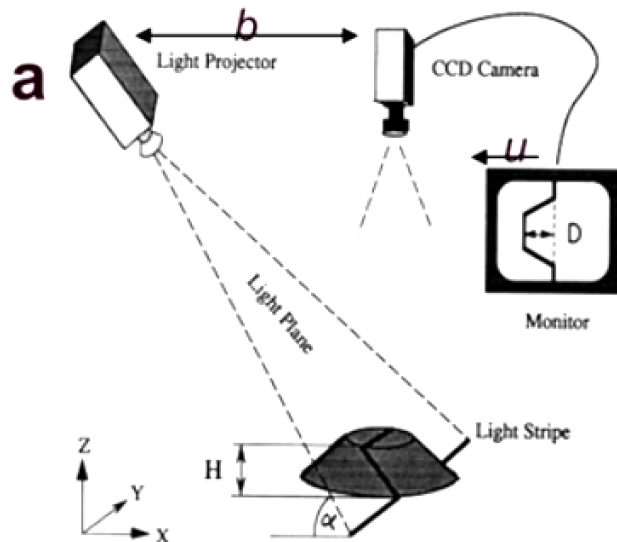
- Sonars offer a good compromise between cost, accuracy, power consumption and ease of integration.
- They are suitable for the detection of obstacles, but because of their low readings rate, they are not very suitable for mapping and location, but fine for obstacle avoidance.
- The robotic application processing sonar data should validate the data, discarding inconsistent measures and integrating multiple measures.
- Recent generation of sonars have a reduced opening angle ($\sim 5^\circ$) and high accuracy (cost around \$ 100.00).

- Sonars are fundamental in educational robotics and Hobby, because of their



Range sensors: structured light

- Eliminates the correspondence problem by projecting a known pattern or structure on the environment, via an emitter.
- a) Systems emits a light texture or collimated light (i.e. light whose rays are parallel, and spread minimally as it propagates), possibly made of laser by means of a rotated mirror.
- b) System projects a laser stripe by turning a laser beam into a plane with a prism.
- The light is perceived by camera
- Range to an illuminated point can then be determined from simple geometry



Range sensors: structured light

- Kinect 1 is a RGBD camera, which is a sensor that integrates a color camera, an infrared camera and an infrared emitter.
- The color camera provides a conventional image while the infrared camera provides a depth map.
- Given a pixel (x, y) in color camera, the same pixel (x, y) in the disparity map provides a value of the distance from the point to the camera.
- The Kinect 1 uses a patented technology where the infrared transmitter emits a grid points whose reflected image is processed and converted to depth information.



Range sensors: structured light

■ Typical characteristics (Kinect):

Range: 1 m to 4.5 m

Resolution: 1.5 to 50 mm.

Image Size:

- color cam: 640x480 (Xbox), 1280x960 (Windows)
- Depth cam: 320x240

Viewing angle: 43° vertical by 57° horizontal

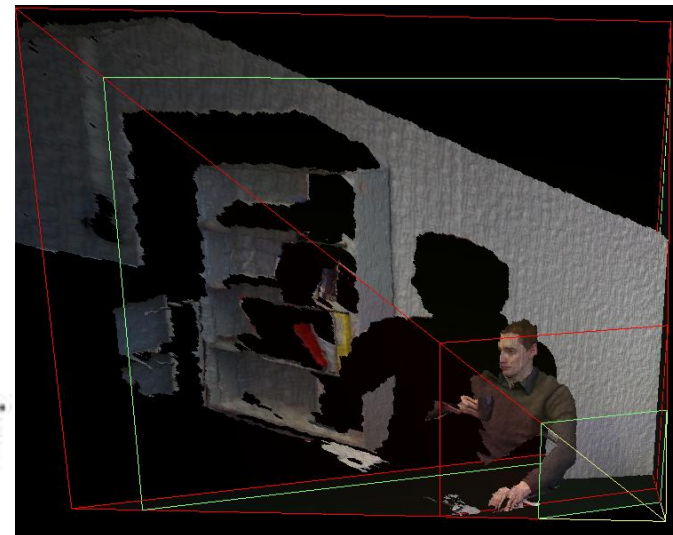
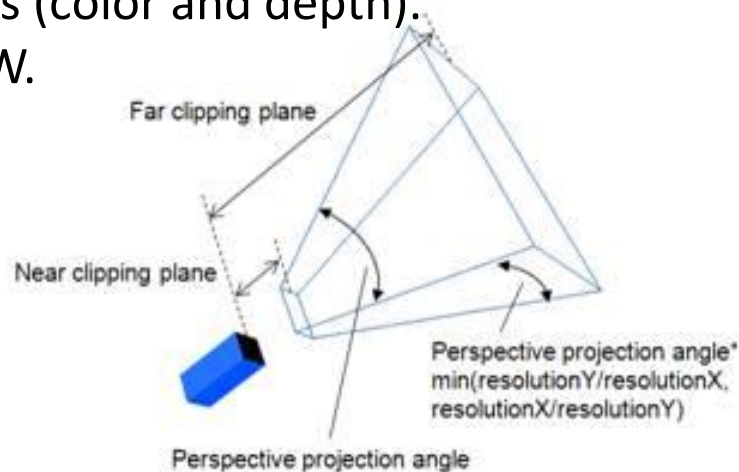
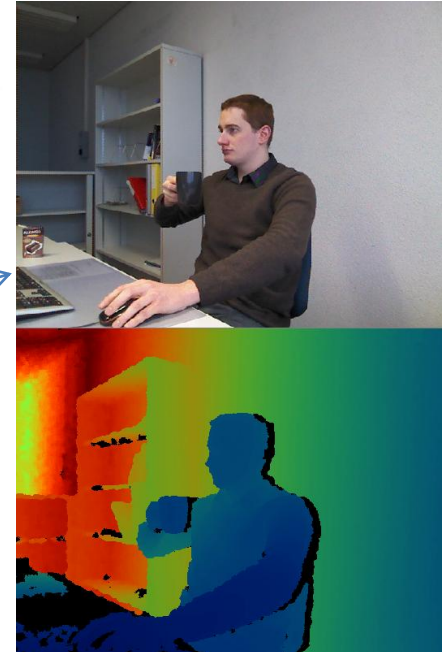
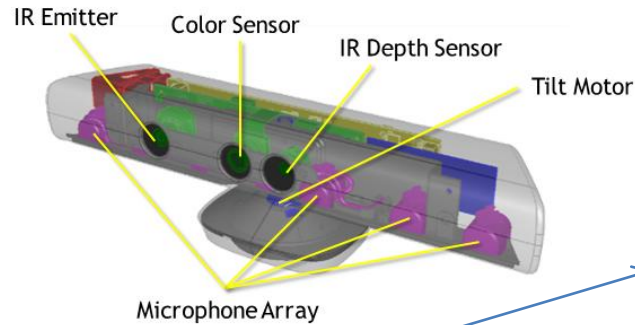
Image Type: color and monochrome (depth).

Sensor: CMOS

capture rate: 30 fps (color and depth).

Consumption: 12 W.

Connectivity: USB.



Range sensors: structured light

■ Depth image to Point cloud:

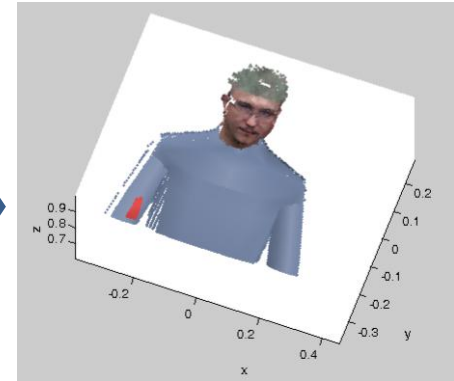
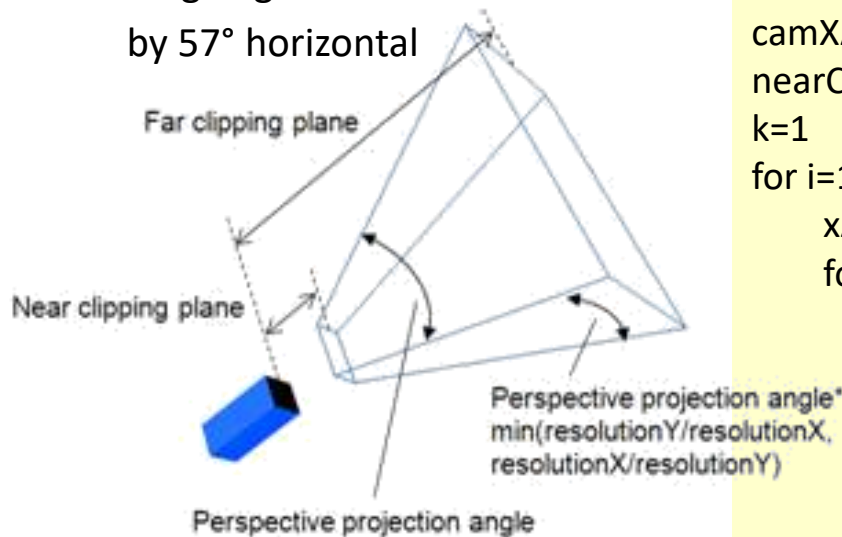
Range: 0.4 m to 4.5 m

- near clipping plane = 0.4m
- Far clipping plane = 4.5m

Image size:

- color cam: 640x480 => max RGB Vision Sensor resolution
- Depth cam: 320x240 => max Depth Vision sensor resolution

Viewing angle: 43° vertical
by 57° horizontal



```
camXResolution=640, camYResolution=480
camXAngle=57, camYAngle= 43 (i.e. 57 *480/640)
nearClippingPlane=0.4 , depthAmplitude=4.5-0.4= 4.1
k=1
```

```
for i=1 to 640 do
```

```
    xAngle=((320-i-0.5)/320)*camXAngle*1/2
```

```
    for j=1 to 480 do
```

```
        yAngle=((j-240+0.5)/240)*camY*1/2
```

```
        depthValue=depthBuffer[i+(j-1)*640]
```

```
        zCoord[k] = nearClippingPlane+depthAmplitude*depthValue
```

```
        xCoord[k] = tan(xAngle)*zCoord[k]
```

```
        yCoord[k] = tan(yAngle)*zCoord[k]
```

```
        k=k+1
```

```
    end
```

```
end
```

Range sensors: structured light

- Kinect and other sensors with infrared only work indoors.
- These sensors are low cost distance sensor but also are used for Human Machine Interface and body posture estimation
- The Kinect 1 also has a vertical positioner (tilt) highly inaccurate and 2 microphones.
- The software infrastructure are more focused on the development of games such as motion detection and posture (tracking skeleton), but also provide access to the depth of the matrix. The main infrastructures are:

Kinect for Windows SDK - only for Windows systems.

OpenNI - multiplatform (Windows, Linux) - supports Kinect and Asus Xtion.

OpenKinect - Linux (support only for XBox version of the Kinect).

Range sensors: Time of flight's cameras

- Swiss ranger (discontinued)
- The SR4000 series Time-of-Flight (ToF) cameras are optical imaging systems which provide real time distance data at video frame rates in a compact, industrial grade design.

Illumination wavelength	850 nm
Maximum frame rate	50 fps
Pixel array size	QCIF – 176 (h) x 144 (v)
Field of view	43.6° (h) x 34.6° (v)
Pixel pitch	40 μ m
Voltage	12.0 V
Current	0.8 A nominal, 1.0A maximum



V-REP demo

Range sensors: Time of flight's cameras

■ Kinect 2

Kinect For Windows 1



Processed Image From Kinect

Kinect For Windows 2



Image via <http://blogs.msdn.com>

Feature	Kinect for Windows 1	Kinect for Windows 2
Color Camera	640 x 480 @30 fps	1920 x 1080 @30 fps
Depth Camera	320 x 240	512 x 424
Max Depth Distance	~4.5 M	~4.5 M
Min Depth Distance	40 cm in near mode	50 cm
Horizontal Field of View	57 degrees	70 degrees
Vertical Field of View	43 degrees	60 degrees
Tilt Motor	yes	no
Skeleton Joints Defined	20 joints	26 joints
Full Skeletons Tracked	2	6
USB Standard	2.0	3.0
Supported OS	Win 7, Win 8	Win 8
Price	\$299	TBD



Range sensors: Laser Range Finders

■ Typical specs:

Measuring range: 5 cm to 20 m (indoor) - 80 m (outdoor).

Inaccuracy: 20-60 mm.

Field of view (scan): 180, 270, 360 ° at intervals of 0.25, 0.5 or 1.0°.

Wavelength: 850-900 nm.

Power: Class I (eye-safe)

Scan rate: 15 to 50 scans / s.

Power supply: 5 V, 10-28 V

Consumption: 3-22 W.

Connectivity: USB, RS-232/422, CAN-BUS, Ethernet / TCP-IP.



Sick LMS-100 (US\$ 5.700,00)



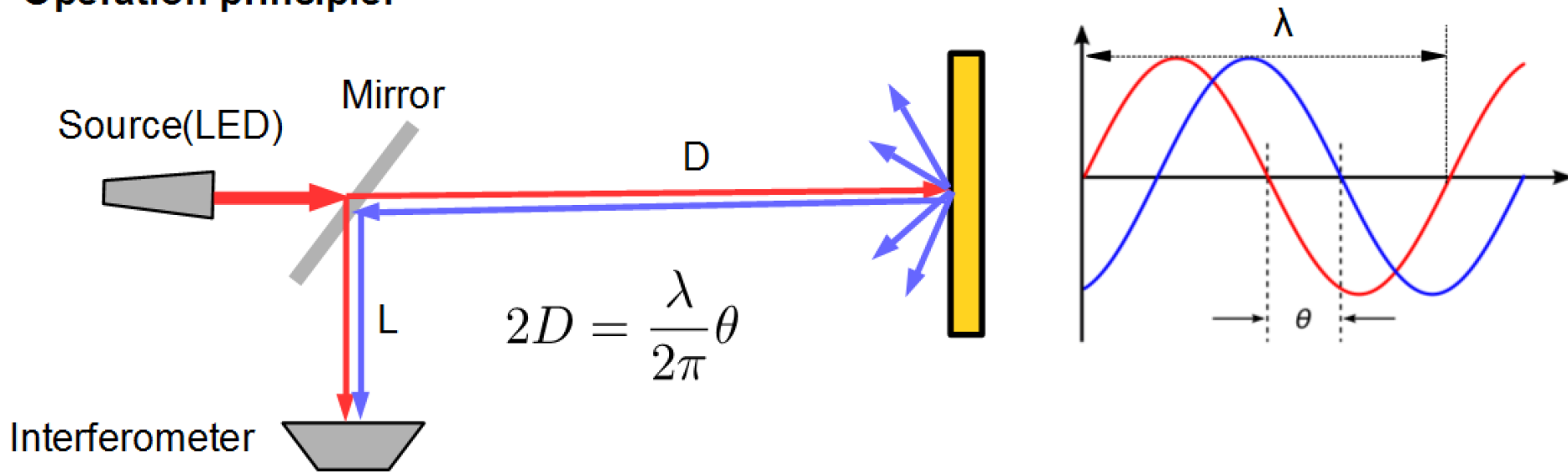
Sick TIM-551 (US\$ 2.500,00)



Hokuyo URG (US\$ 1.150,00)

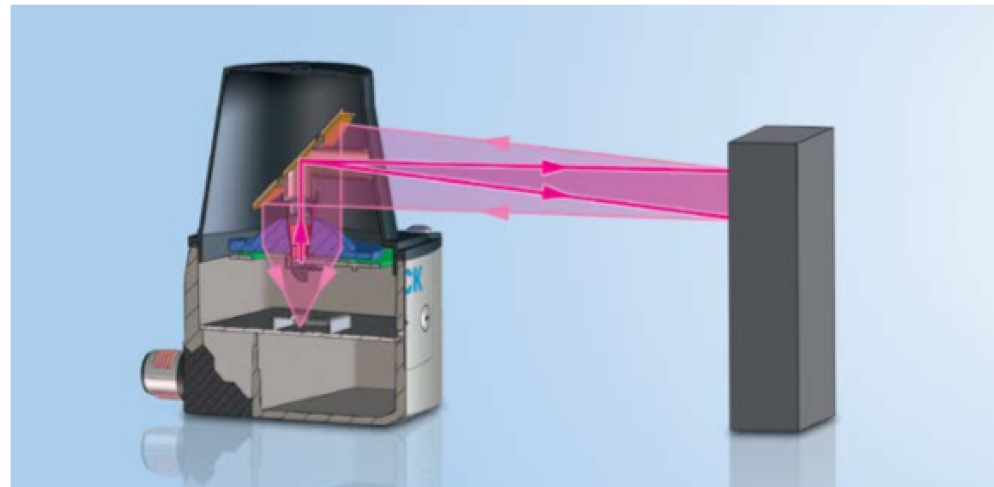
Range sensors: Laser Range Finders

Operation principle:



The ~ 900 nm light is modulated at a frequency f that produces wavelength $\lambda = c / f$. Example: $f = 5$ MHz $\rightarrow \lambda = 60$ m.

For $D = 5$ m the interferometer should detect a phase difference (θ) of ~ 1 rad.

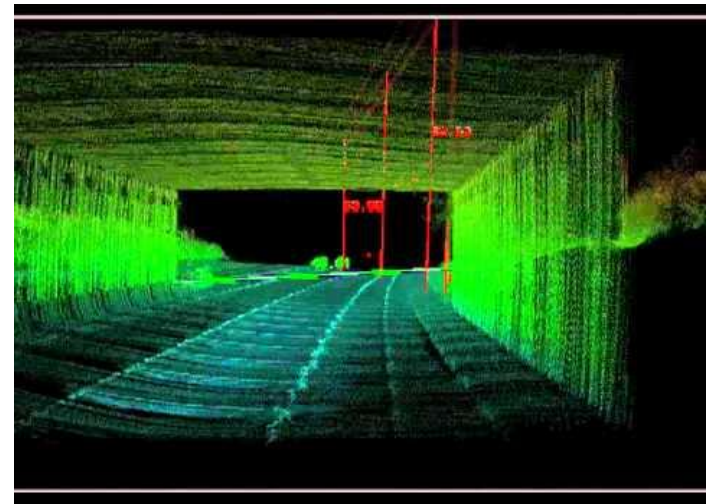


Range sensors: Laser Range Finders

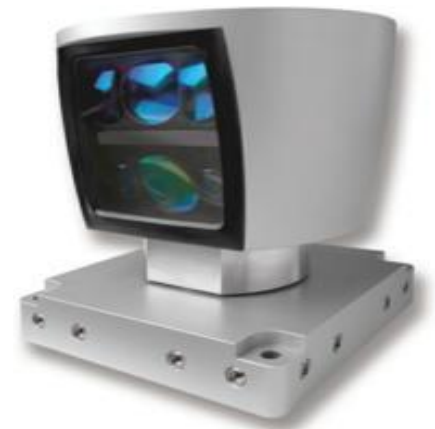
- laser distance sensors are standard in robotic applications since they allow not only to detect obstacles with high accuracy but also infer form (important for the detection of environmental features such as corners of walls).
- They are very used sensors for mapping and precise location.
- The major limitations are cost (but soon will become cheap) and the necessity to have great processing capability to extract information provided by the high volume of data points of the sensor.
- When mounted in positioning units (pan-tilt) produce 3D environment mapping (point clouds).

Range sensors: Laser Range Finders 3D

- There are sensors with 3D scanning capability. They consist of arrangements with groups of transmitters and receivers mounted on the turntable. They are mainly used in autonomous cars like Google Car.



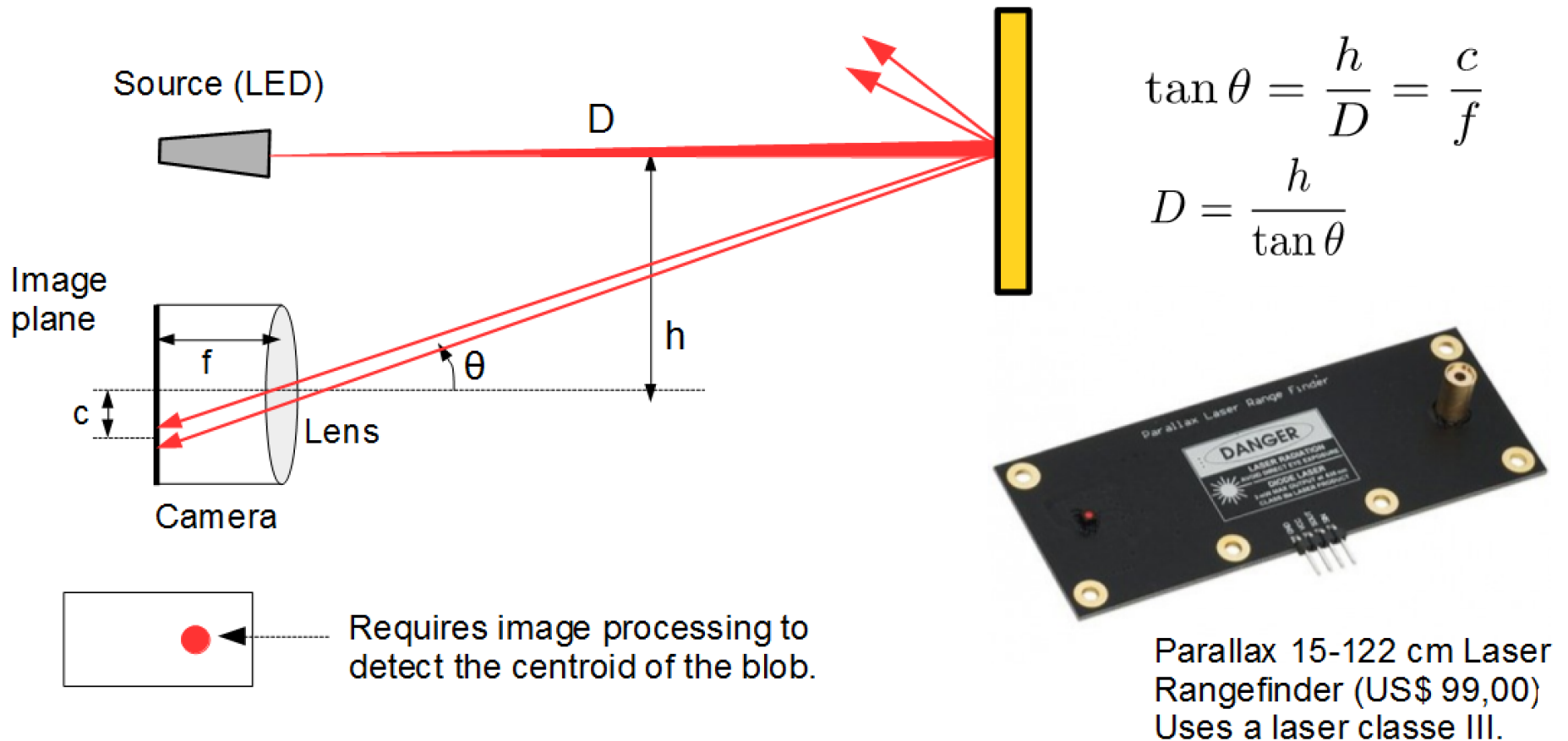
- Velodyne HDL-64E (US \$ 75,000.00)
Horizontal opening: 360° with 0.08° resolution.
Vertical opening: 26,8°.
Scan rate: 1.3×10^3 dots / s.
Maximum range: 50 m (asphalt), 120 m (tall objects).
Imprecision: 2cm.



Demo V-REP

Range sensors: Laser Range Finders 1D

- Measure the distance in one direction by combining an LED emitter and a CCD sensor (camera). More accurate (and expensive) than infrared sensor. More directional than a sonar.

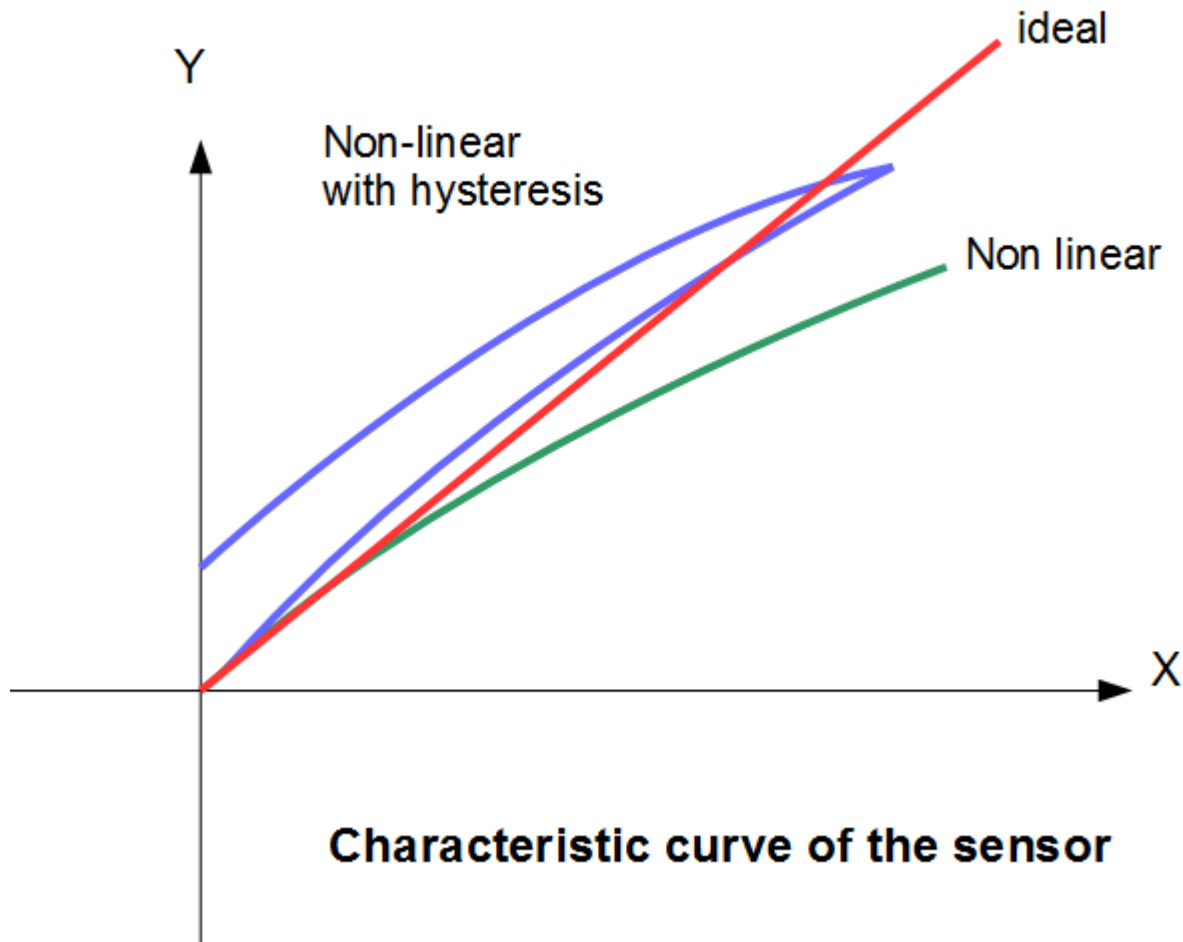


Sensors characterization and performances

See: <http://www.ni.com/white-paper/14860/en/>

- Sensor input (X): physical quantity that activates the sensor. Example: angular rotation, acceleration, reflected light, etc.
- Sensor output (Y): physical quantity measured by the sensor. Example: angular speed, acceleration, distance, etc.
- Characteristic curve of the sensor: Y x X curve

Sensors characterization and performances



Sensors characterization and performances

- How to quantitatively determine the sensor's performance?
Important for the correct use of sensor measurements in robotic applications.
- Characteristics obtained in laboratory (commonly provided in the data sheets of the sensors):
 - Dynamic Range (range);
 - Resolution;
 - Linearity / non-linearity;
 - Band or frequency;
 - Response Time.
- Characteristics obtained in real applications (in situ)
 - Sensitivity;
 - Cross sensitivity;
 - Accuracy;
 - Errors;
 - Precision

Sensors characterization and performances

- **Dynamic Range** (range): is the ratio (or its square) between the highest and the lowest sensor output, usually in dB.

Example: A sonar with a band of 5 cm - 5m has dynamic range of:

$$10\log \frac{500}{5} = 20dB$$

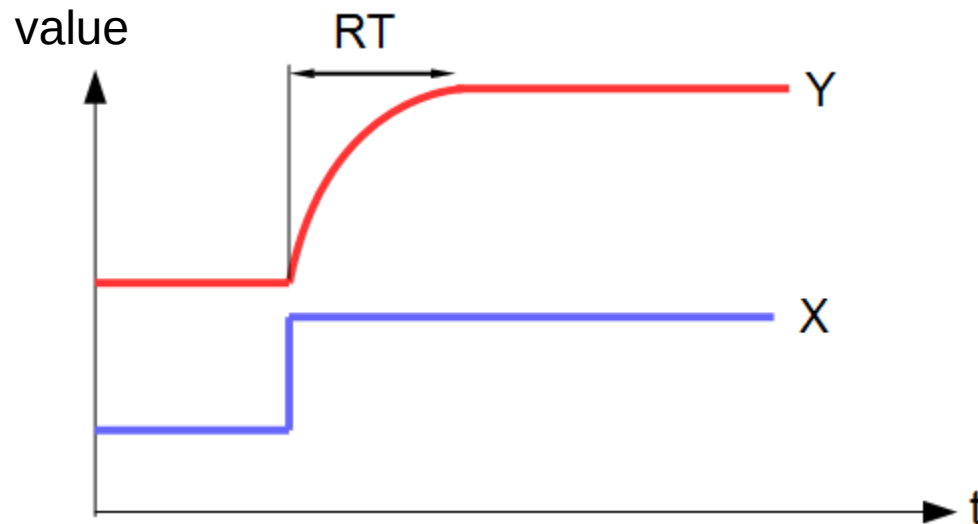
- **Resolution:** is the smallest increment in the input that produces a change in the sensor's output. It depends of the A/D converter (bits used in the digitalization) and of the sensor accuracy (e.g., number of pixels of a CMOS sensor).

Example: The analog port of an Arduino microcontroller (0-5V) has a 10 bit A/D converter, which results in a resolution of:

$$\frac{5}{2^{10}} = 4.89mV$$

Sensors characterization and performances

- **Frequency:** is the rate of measurement that we can get out of a sensor.
Example: 50 measurements / sec (50 Hz) for a modern distance measuring laser sensor.
- **Response time (TR):** the time the sensor takes to change the output from a change in the input

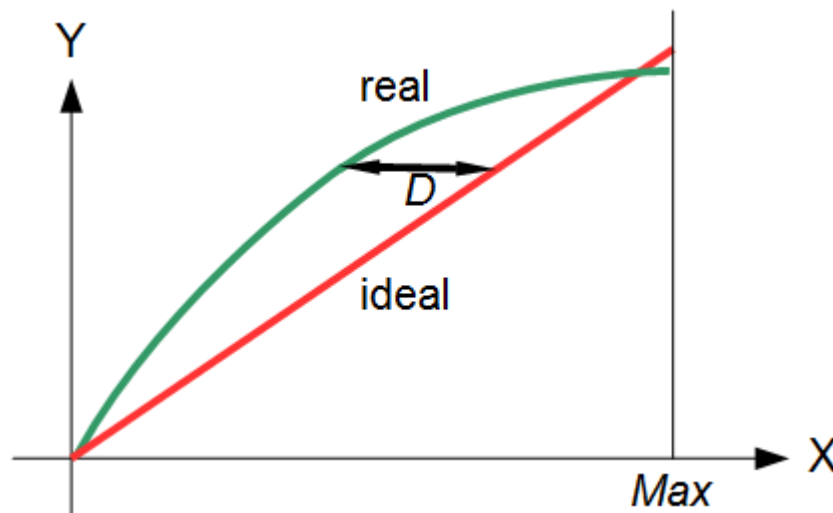


Sensors characterization and performances

- **Linearity**: is a measure of how much the characteristic curve of the sensor approaches the ideal curve.

Example: the gyroscope ITG3200 has a range of 14.375 in the measures it provides. The sensor is considered to be linear when producing readings approaching the function $y = 14.375 \omega$.

- **Non-linearity (NL)**: how much (%) a characteristic curve deviates from the ideal curve.



$$NL = \frac{D}{\max} \times 100\%$$

Sensors characterization and performances

- **Sensitivity**: is the smallest absolute amount of change that can be detected by a measurement (how sensitive the sensor is). It is given by the slope of the characteristic curve of the sensor.
- **Accuracy** (ac): is how close the measurements provided by the sensor are from the true values v .

$$ac = 1 - \frac{|error|}{v}$$

Sensors characterization and performances

- **Error:** is the difference between the value provided by the sensor (m) and the true value (v) real of the measure of magnitude.
- **Precision (p):** is the sensor's ability to always provide the same value for a measured quantity that does not change (not synonymous of accuracy). If the sensor shows non-systematic Gaussian error with a standard deviation σ , the precision is defined as:

$$p = \frac{\text{range}}{\sigma}$$

Uncertainty Representation

- Sensing is always related to uncertainties
 - How can uncertainty be represented or quantified?
 - How do they propagate – uncertainty of a function of uncertain values?
- **Systematic errors:** are deterministic (predictable), and can therefore be measured and compensated (via calibration)
for example: the diameter of the robot wheels, the distance between the wheels, stereo camera bended after a shock...
- **Random errors:** they occur randomly, cannot be mitigated by more precise sensor machinery and can be modeled only with stochastic techniques.(i.e. laser range finder reading=> activity 3)
for example: Hue instability in a color camera, spurious range finding errors, black level noise in a camera...

Uncertainty Representation

From a statistical point of view we wish to characterize the error of a sensor, not for on specific measurement, but for any measurement-> the problem of sensing becomes an estimation problem where the goal is to characterize the estimate of the true value given N measurements.

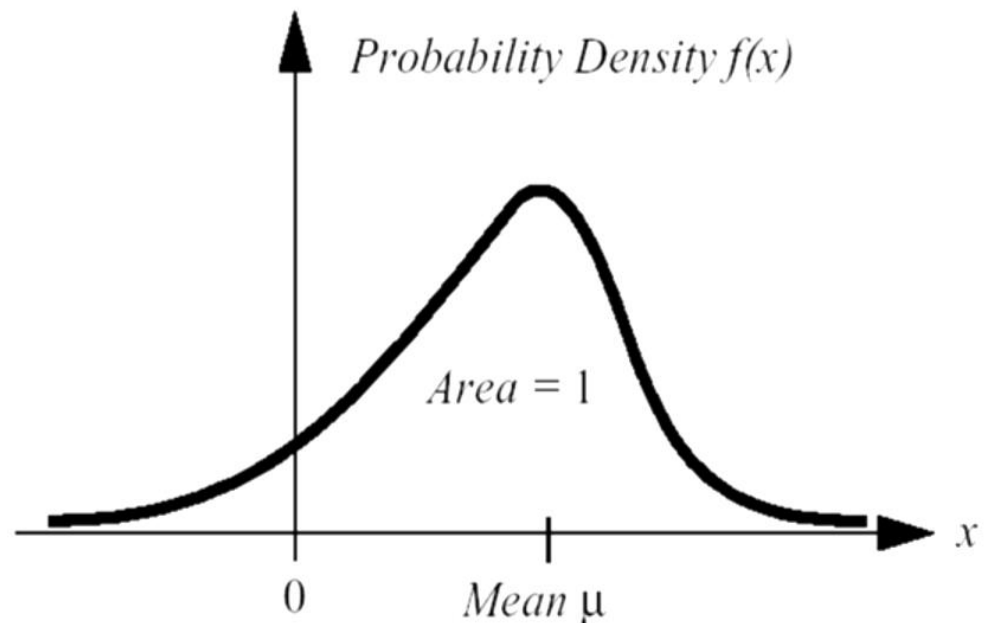
The true value is represented by a random unknown variable x . We use a probability function f to characterize the statistical properties of the value of x . using this function we can quantitatively define the mean μ , the variance σ^2 and the standard deviation σ

- The density function identifies for each possible x value of X , a probability function $f(x)$ density along the y - axis
- The area under the curve is 1, indicating the complete chance of X having some value

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

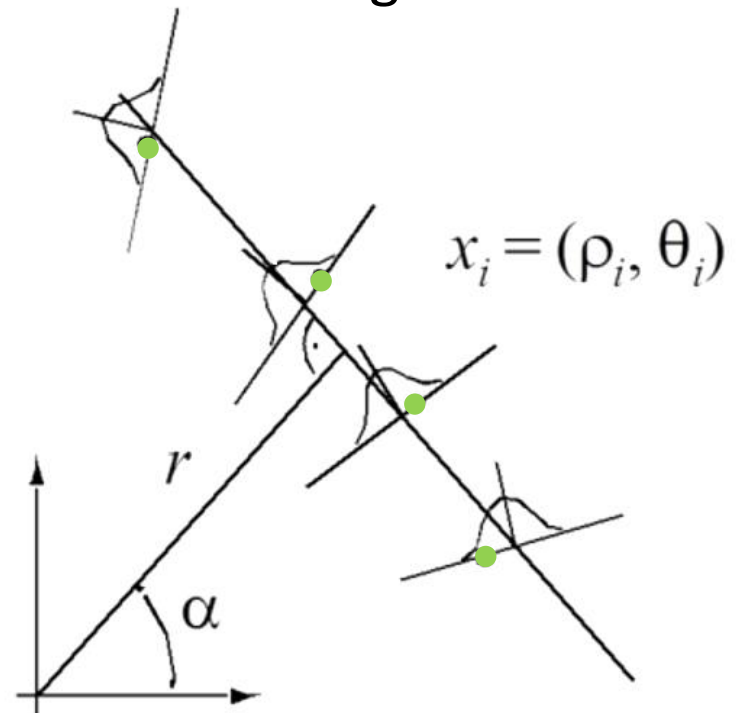
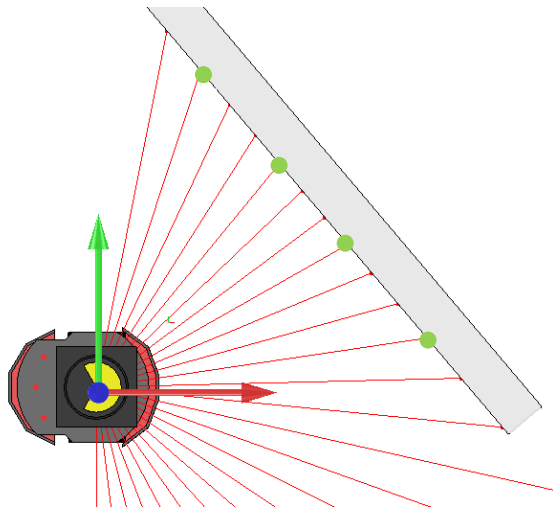
$$\mu = E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$



Uncertainty Representation: error propagation law

- Imagine extracting a line based on point measurements with uncertainties
- The model parameter ρ_i (length of the perpendicular) and θ_i (its angle to the abscissa) describe a line uniquely
- What is the uncertainty of the extracted line knowing the uncertainties of the measurement points that contribute to it?



Uncertainty Representation: error propagation law

- Error propagation in a multiple-input multi-output system with n inputs and m outputs $Y_j = f_j(X_1, \dots, X_n)$

What can we say about the uncertainty of Y_j based on the knowledge of the uncertainty X_i ?

i.e. How to propagate the error from the inputs to the output to discover the uncertainty in the variable Y ?

The general solution can be generated using first order Taylor expansion of f

- It can be shown that the output covariance matrix C_Y is given by the error propagation law: $C_Y = F_X C_X F_X^T$

- C_X : covariance matrix representing the input uncertainties

- C_Y : covariance matrix representing the propagated uncertainties for the outputs

- F_X : is the Jacobian matrix defined as

$$F_X = \begin{bmatrix} \frac{\partial f_1}{\partial X_1} & \dots & \frac{\partial f_1}{\partial X_n} \\ \vdots & \dots & \vdots \\ \frac{\partial f_m}{\partial X_1} & \dots & \frac{\partial f_m}{\partial X_n} \end{bmatrix}$$